Active Cartesian Modular Dynamics: Machine-Verified Foundations

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Repository: https://github.com/jeffgurd888/thet-logos

> "If we don't stop worrying about the Almighty dollar, we're not going to change the world!" — Caleb Gurd, as remembered by his father.

Version note. Version 1 (2026-10-15; DOI: 10.5281/zenodo.23408680) reported the program through Rung 8, together with the SM-Match program and the first round of the theories intake. Version 2 (2026-10-16; DOI: 10.5281/zenodo.23427964) added the closing of Rung 8's bosonic half (Rung 8B, §8B), fermionic second quantization (Rung 9, §9), bosonic second quantization (Rung 10, §10), and a fenced background-algebra appendix (Appendix A). Version 3 adds: the infinite-mode free theory (Pitch 2, §10A); interacting Hamiltonians at finite modes for both statistics, with the Pitch 3 verdict stated exactly as ruled (§10B); finite-dimensional Schrödinger dynamics (§10C); the discharge of Rung 7's boost window (§7 amendment); the graph-Laplacian bridge exhibit (Appendix A); and the opening stages of the continuum pitch (Pitch 4, §13A) — the lattice scoping, Stage 0, Stage 1a, and Stage 1b in its four sub-stages. It also carries a full re-audit of §16 (what is NOT claimed): two lines carried from Version 2 were overtaken by the work and are amended in the open (§0, corrections to Version 2). Everything else is carried from Version 2 unchanged except as amended in the open where dated amendment notes say so. Version 3 is published as a new version of the Version 2 record; its DOI is minted on publication.

Abstract

This paper states exactly what the machine has verified in the Active Cartesian Modular Dynamics (ACMD) program through Pitch 4, Stage 1b, together with the SM-Match program and the first round of the theories intake. Every theorem named here is proved in Lean 4 (tier T1), explicitly axiomatized with proved consequences (T2), stated as a labeled open target (T3/T5), or recorded as refuted. The program is machine-verified mathematical scaffolding toward a rigorous four-dimensional quantum field theory: a finite spectral triple and its classification; the Connes–Kreimer Hopf algebra of renormalization; the almost-commutative product with its twist derived rather than assumed; analytic and Hilbert-space infrastructure for a concrete model; verdicts on all six Wightman axioms in a one-mode fermionic model; genuine Schwartz test-function machinery and a geometric Poincaré action; a finite-dimensional Stone theorem and a joint functional calculus for commuting symmetric operators; a Fock-space inner product with proved creation/annihilation adjoints on both the fermionic and bosonic sides; fermionic second quantization — the functorial

lift Γ(U), the differential lift dΓ, and a free Hamiltonian whose gap is derived at finite modes; the bosonic dΓ lift with its own derived free gap, and the bosonic Γ(U) lift, completed as a verified functor by the same-day campaign that proved the two permanent identities its functoriality had been priced at; a structural match against the Standard Model's gauge algebra, fermion content, classical Lagrangian sectors, and Higgs breaking pattern; and fit verdicts for KMS theory, classical Yang–Mills, Haag–Kastler algebraic QFT, and the finite spectral action. Version 3 adds: the free theory at countably many modes for both statistics, with the derived free gap in the eigenvalue-on-finite-particle-domain sense (Pitch 2); interacting Hamiltonians at finite modes — the fermionic campaign stands whole with gap persistence, the bosonic campaign stands except for essential self-adjointness, its sole priced clause (Pitch 3); unitary Schrödinger dynamics with picture equivalence and proved time-invariance of the interacting gap clusters; full restricted-Lorentz invariance of the free one-particle mass-shell measure (the Rung 7 boost window, discharged); and the opening of the lattice road toward the continuum (Pitch 4): Wilson-action lattice gauge theory with exact gauge invariance, the Gibbs measure and correlation package, Elitzur's theorem at a general gauge group and every coupling, the reflection structure, the Osterwalder–Schrader form with positivity proved at zero coupling and reduced at general coupling to one named lemma, the OS Hilbert space, and the transfer matrix — with the lattice gap proved exactly at zero coupling and fenced, with its price, at general coupling.

What this is not. No continuum limit is claimed. No interacting 4D quantum field theory is claimed. Interacting Hamiltonians are constructed (§10B) — at finite modes, for matter self-interaction only, with no gauge field; no interacting continuum theory of either statistics is constructed, and no scattering content appears anywhere. The Yang–Mills existence and mass gap problem is the program's stated horizon and mission target; it is not claimed, approached in the Clay sense, or implied by anything below. The mass gap appears in this paper only as a fixed, quarantined target statement against which future partial results can be measured; the gaps that are derived below (§§9, 10, 10A, and the lattice gap of §13A) are gaps of free Hamiltonians at finite or countably many modes, of an interacting finite-mode Hamiltonian under a proved persistence threshold, and of a finite-lattice transfer matrix at zero coupling — labeled as such wherever they occur. **Parts of the official Clay Yang–Mills statement closed by this corpus: 0.**

Tier key. T1: proved in Lean 4, zero `sorry`s. T2: axiomatized, with proved (T1) consequences. T3: theorem stated, proof open, labeled in-file. T5: opaque, quarantined, or priced window — theorem-shaped, content not established. KILL: refuted or withdrawn; returns only as a new entry by proof.

Verification status. All verification claims for Rungs 1–8 were re-checked on 2026-10-15 (§0). That re-check raised two flags — the finite spectral-action proofs failed to compile under the pinned toolchain (§12) and the Morley gauntlet file could not be rebuilt to completion in that pass (§13) — and **both flags were resolved on 2026-10-15**: `SpectralActionFinite.lean` was repaired (proof-local fixes only; no statement changed; no `sorry`/`admit`/axiom added) and its FIT verdict stands restored (§12), and `EdgeSignature.lean` was re-verified by a standalone green build (exit 0; zero

`sorry`/`admit`; axiom checks on the 13 headline theorems clean; §13). The material added in Version 2 (§§8B, 9, 10, Appendix A) was verified at its landings on 2026-10-15/16 as reported in the §0 addendum. The material added in Version 3 (§§10A, 10B, 10C, 13A, the §7 amendment, and the Appendix A bridge exhibit) was verified at its landings on 2026-10-16/17 as reported in the §0 Version 3 addendum, where every count in this paper is re-derived from the current sources by a single stated method.

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0. Verification protocol and corpus inventory

Counting method (uniform, reproducible). All counts in this paper are counts of `theorem` and `lemma` declarations, obtained by the regular expression `^\s*(private\s+|protected\s+)?(theorem|lemma)\s` applied to every file in `lean/ThetLogos/`. Counts include auxiliary lemmas; they are counts of declarations, not of "importance." **Inventory freeze:** the corpus was inventoried at commit `790c1e0` on 2026-10-15. Work landing after the freeze (a Rung 9 start and a ternary substrate module, both in progress under separate directives) is outside this paper's scope and is not assessed here.

Corpus totals (exact).

Scope	Files	Theorem/lemma declarations
Whole Lean corpus (`lean/ThetLogos/`)	77	**2,016**
ACMD program files (Rungs 1–8, SM-Match, intake, Morley, targets/dossiers)	64	**1,352**
Earlier exploratory campaigns in the same repository (CF-kernel arc, exotic sectors, chromatic and related work)	13	664

The figure "300+" that circulated in earlier drafts of program summaries was never derived from a count of this corpus. It is withdrawn and replaced by the exact counts above.

Build (2026-10-15, reported exactly). `cd lean && lake build` completes successfully (3,324 jobs). Stated plainly: the root module `ThetLogos.lean` imports only 14 core modules, so the remaining 63 modules were additionally built one by one (`lake build ThetLogos.<Module>`). Result: **74 of 77 modules build green.** The three exceptions:

- `SpectralActionFinite.lean` **failed** at the 2026-10-15 re-check: four proof errors (`simp` made no progress / failed rewrite at lines 942, 1006, 1024, 1074), all inside the a_2 trace-computation proofs and consistent with Mathlib deprecation drift since the file's 2026-10-11 verification. No statement was changed and no `sorry` introduced; the proofs no longer closed under the pinned toolchain. **Repaired the same day (2026-10-15): proof-local fixes only; the file**

builds green again and the FIT verdict of §12 stands restored (theories-intake dossier amended accordingly).

- `SpectralActionBridge.lean` failed transitively (it imports the file above); restored with the repair.
- `EdgeSignature.lean` (the Morley gauntlet file, §13): two rebuild attempts on 2026-10-15 timed out (10 and 25 minutes, under heavy machine contention) without emitting any error — the build did not finish in that pass. **Re-verified later on 2026-10-15 by a standalone build: exit 0, zero `sorry`/`admit`, axiom checks on the 13 headline theorems clean. Closed.**

A minor wart, recorded for completeness: two declaration names collide across modules in the shared `ThetLogos` namespace (`vacuum_cyclic` in `FreeField` and `FermionicFock`; `Dprod` in `ProductTriple` and `CausalProduct`), so those module pairs cannot be imported into a single environment. Each module builds and verifies on its own. Wiring the newer modules into the root import list, and repairing the spectral-action proofs, are queued mechanical follow-ups.

Sorry hygiene (exact). The corpus contains exactly **seven** `sorry`s, all pre-existing, all labeled in-file, none introduced by Rungs 1–8, SM-Match, or the intake:

Location	Count	Tier	Content
`BlockedQuestions.lean` (lines 76, 108, 140, 194, 274)	5	T5	Open conjectures Q1–Q5, filed as questions
`ModularTime.lean:206`	1	T3	`kms_identity_diagonal`; numerically verified, proof open
`XYZTripotents.lean:492`	1	T5	`xyz_tripotents_exist`; single documented target

There is no `admit` anywhere in the corpus. Every other occurrence of the word "sorry" in the sources is documentation prose, mostly statements that a file contains none.

Axiom footprint (spot-checked 2026-10-15). `#print axioms` was run on thirteen headline results. Eleven depend only on Lean's standard axioms `[propext, Classical.choice, Quot.sound]` — no `sorryAx`: `antipodeAxiomEval_proved`, `modularFlux_selfAdjoint`, `w5_span_true`, `wick_four_point_operator`, `schwartzComplete_discharge`, `jointFC_spectralMapping`, `higgs_stabilizer_is_Q`, `gibbs_kms_weight`, `stone_finiteDim`, `ann_adjoint`, `adag_adjoint`. The two labeled exceptions behave exactly as their labels say: `pauliBounded` depends additionally on the one T2 axiom `pauli_compactResolvent`, and `trace_collapse_hypothesis_fails` on the `CausalProduct` file-level T2 product axioms. And `kms_identity_diagonal` depends on `sorryAx` — precisely its T3 label.

v2 addendum (2026-10-16): increments since the Version 1 freeze

The Version 1 freeze above stands as the 2026-10-15 record. The work that landed after it — and that Version 2 reports — is inventoried here by the same counting method, as increments.

Program total (Version 2, decomposed). Version 2's program total, by the same decomposed method as the §0 table, is **1,648**

Increment	**Files**	**Theorem/lemma declarations (definitions counted separately)**
Rungs 1–8 era (Version 1 freeze)	64 program files	1,352
Rung 9 — fermionic second quantization (§9)	2	+50 (plus 13 definitions)
Rung 8B — bosonic assembly (§8B)	1	+31 (plus 6 definitions)
Rung 10 — bosonic second quantization (§10)	2	+31 (plus 13 definitions)
Background algebra — sexagesimal clock (Appendix A)	1	+31 (plus 12 definitions, 1 structure)
Γ functor campaign — Rung 10 Task 3 closed (§10)	1	+42 (plus 8 definitions)
Background algebra — flux-twist toy (Appendix A)	1	+38 (plus 10 definitions)
Background algebra — weld duality (Appendix A)	1	+73 (plus 18 definitions)
Program total (decomposed)		**1,648**

The ternary substrate module of Appendix A (`TritGrading.lean`, 350 lines) also landed after the Version 1 freeze; it is reported qualitatively in Appendix A and carries no separate increment in the decomposed chain above.

Builds (as reported at each landing). The root `lake build` was green (3,325 jobs, exit 0) at the Rung 9, Rung 8B, and Rung 10 landings, with the pre-existing corpus untouched. The newer rung modules are not in `ThetLogos.lean`'s root import list — the same staging as at Version 1 — and are verified by targeted builds, each green: `ThetLogos.SecondQuantGamma` and `ThetLogos.SecondQuantDGamma` (Rung 9, within the root build); `ThetLogos.BosonicAssembly` (2,452 jobs); `ThetLogos.BosonicSecondQuant` (2,453 jobs); `ThetLogos.BosonicGamma` (2,454 jobs); `ThetLogos.SexagesimalClock` (3,118 jobs, zero errors, zero warnings); `ThetLogos.FluxTwist` (3,120 jobs); `ThetLogos.WeldDuality` (3,120 jobs). Wiring the newer modules into the root import list remains a separate, queued decision.

Sorry hygiene (increments). The new files introduce no `sorry` and no `admit`: zero-sorry sweeps are reported clean for both Rung 9 files, for `BosonicAssembly.lean`, for both Rung 10 files, for `TritGrading.lean`, for `SexagesimalClock.lean` (which also has no `native_decide`), and for `FluxTwist.lean` and `WeldDuality.lean` (zero `sorry`/`admit`/`native_decide` in code for both). The corpus total of §0 — exactly seven, all pre-existing and labeled — is unchanged.

Axiom footprint (increments). `#print axioms` spot-checks on the new headline results return exactly `[propext, Classical.choice, Quot.sound]` — no `sorryAx`, no new

axioms: every headline theorem of both Rung 9 files (per that rung's dossier); eleven named headlines of Rung 8B, including `adjoint_relation`, `ccr_mixed`, `numberOp_basis`, `sector_orthogonal`, and `cycSub_eq_top`; fifteen named headlines of Rung 10 across its three tasks, including `dgammaBos_adjoint`, `freeHamiltonianBos_eigenvalue_gap`, and `gammaBos_apply_basis`; and eight named headlines of the sexagesimal clock module, including `crtEquiv` and `ClockGrading.reduceToTrit`; six named headlines of the flux-twist module, including `twistedShift_unitary` and `twistedShift_mul_gaugeMatrix`; and seven named headlines of the weld module, including `trace_weldPlus`, `etaCorr_sq`, and `signature_etaDirac`. The clock module's `lcm_one_through_six` depends on no axioms at all.

v3 addendum (2026-10-17): increments since the Version 2 freeze, counted by one method

The Version 2 record above stands as published. The work that landed after it — and that Version 3 reports — is inventoried here. For this addendum every count was **re-derived from the current sources** (at commit `4380a2a`) by a single method, stated exactly so any reader can reproduce it:

Counting method (Version 3 — strict line-initial, retained). A theorem or lemma is counted when a line of the comment-stripped source begins at column 0 with `theorem` or `lemma`: the regular expression `^(theorem|lemma)` applied after stripping Lean comments (line and nested block comments). This is the Version 2 §0 pattern, retained unchanged by the author's decision of 2026-10-17, taken after both candidate rules had been computed and compared (see the recount note below): the strict rule is the one whose total any reader can reproduce with the simplest possible grep, and it never overcounts. Definitions are counted by the analogous line-initial pattern over `def`/`abbrev`/`structure`/`instance`, admitting the standard same-line prefixes (an attribute, `noncomputable`, `private`, `protected`) that definition keywords routinely carry; the program total itself is the theorem/lemma count. The recount consequences follow, recorded here rather than absorbed silently:

- **The Version 2 total stands.** Recounting the Version 2 scope by the strict rule gives exactly its published **1,648**, and the Rung 9 increment is 50 as published in §0. Fifteen declarations in today's program modules carry a same-line `@[simp]` attribute prefix and are therefore **not** counted by the strict rule; an attribute-inclusive count of the same modules totals 2,432. The fifteen, all present and verified since their files landed: `gamma_apply` (Rung 9, §9); `loopEnd_nil`, `loopEnd_cons`, `loopProd_nil`, `loopProd_cons` (Pitch 4 Stage 0 — its increment is 68, as its landing dossier stated); `fermConfigEmb_apply`, `fermConfigEmbLim_apply`, `configEmb_apply`, `configEmbLim_apply` (Pitch 2 Campaign A — its increment is 57 line-initial, not the attribute-inclusive 61 of its dossier); and `boostLin_apply_zero`, `boostLin_apply_one`, `boostLin_apply_two`, `boostLin_apply_three`, `boostLin_symm_apply_zero`, `boostLin_symm_apply_one` (the Rung 7 boost discharge — 80 line-initial, not the attribute-inclusive 86 of its dossier).
- **The running chain is corrected.** The decomposed chain published in the repository's README at the time of assembly totaled 2,427, because it summed

the landing dossiers' figures under their two historical patterns: the Pitch 2 and MassShellBoost dossiers counted attribute-inclusive (+4 and +6 over strict), the rest counted strict. Under the single strict rule the program total is **2,417**, and Version 3 reports that figure everywhere.

Program total (Version 3, decomposed).

Increment	**Files**	**Theorem/lemma declarations (definitions counted separately)**
Rungs 1–8 era (Version 1 freeze)	64 program files	1,352
Rung 9 — fermionic second quantization (§9)	2	+50 (plus 13 definitions)
Rung 8B — bosonic assembly (§8B)	1	+31 (plus 6 definitions)
Rung 10 — bosonic second quantization (§10)	2	+31 (plus 13 definitions)
Background algebra — sexagesimal clock (Appendix A)	1	+31 (plus 13 definitions and structures)
Γ functor campaign — Rung 10 Task 3 closed (§10)	1	+42 (plus 8 definitions)
Background algebra — flux-twist toy (Appendix A)	1	+38 (plus 10 definitions)
Background algebra — weld duality (Appendix A)	1	+73 (plus 18 definitions)
Subtotal — Version 2 scope (strict recount = as published)		1,648
Pitch 2 — free theory at countably many modes (§10A)	5	+137 (plus 57 definitions)
Pitch 3 Campaign 1 — fermionic interaction (§10B)	1	+58 (plus 15 definitions)
Schrödinger dynamics (§10C)	1	+49 (plus 10 definitions)
Pitch 3 Campaign 2 — bosonic interaction (§10B)	1	+69 (plus 21 definitions)
Rung 7 boost window discharged (§7 amendment)	1	+80 (plus 15 definitions)
Background algebra — bridge exhibit (Appendix A)	1	+29 (plus 7 definitions)
Pitch 4 Stage 0 — finite lattice gauge theory (§13A)	1	+68 (plus 33 definitions and instances)
Pitch 4 Stage 1a — correlations and Elitzur (§13A)	1	+68 (plus 13 definitions)
Pitch 4 Stage 1b-A — reflection structure (§13A)	1	+67 (plus 16 definitions)
	1	+44 (plus 10 definitions)

Increment	Files	Theorem/lemma declarations (definitions counted separately)
Pitch 4 Stage 1b-B — OS form and positivity (§13A)		
Pitch 4 Stage 1b-C — OS Hilbert space (§13A)	1	+42 (plus 17 definitions)
Pitch 4 Stage 1b-D — transfer matrix (§13A)	1	+58 (plus 13 definitions)
Program total (decomposed, strict line-initial method)		**2,417**

As at Version 2, the ternary substrate module (`TritGrading.lean`, 27 declarations) is reported qualitatively in Appendix A and carries no separate increment in this chain. For reference only, the whole Lean corpus as it stands today — 103 files, including the earlier exploratory campaigns and every post-freeze module whether or not it belongs to the program chain — counts **3,086** declarations by the strict rule; that figure is an inventory of the tree, not a program total, and is not a partition of the chain above.

Two raw-grep disclosures, so the counts can be checked without surprise: in `FermionicInteraction.lean` a raw line count gives 59 hits — the 59th is the word "theorem" opening a line inside the §4 header comment, disclosed in that campaign's dossier; the comment-stripped count is 58. In `SchrodingerDynamics.lean` the raw count gives 52 hits, of which exactly 3 are header or comment prose, disclosed in its dossier; the count is 49.

Builds (as reported at each landing, parent-verified). Every Version 3 module was verified by a targeted build (`lake build ThetLogos.<Module>`), each green, at its landing: Pitch 2 Campaign A modules (2,456 jobs) and Campaign B (3,207 jobs across the pitch modules); `ThetLogos.FermionicInteraction` (3,297 jobs); `ThetLogos.SchrodingerDynamics` (3,319 jobs); `ThetLogos.BosonicInteraction` (2,704 jobs targeted); `ThetLogos.MassShellBoost` (3,463 jobs); `ThetLogos.BridgeLaplacian` (3,117 jobs); `ThetLogos.LatticeGauge` (3,390 jobs); `ThetLogos.LatticeCorrelations` (3,391 jobs); `ThetLogos.LatticeReflection` (3,392 jobs); `ThetLogos.LatticePositivity` (3,393 jobs); `ThetLogos.LatticeOSSpace` (3,550 jobs); `ThetLogos.LatticeTransfer` (3,551 jobs). The newer modules are not in `ThetLogos.lean`'s root import list — the same staging as at Versions 1–2, disclosed in each module header; wiring them in remains a separate, queued decision.

Sorry hygiene (re-derived 2026-10-17). A comment-stripped sweep of all 103 corpus files finds the word `sorry` in code **exactly seven times** — the same seven labeled occurrences of the §0 table (five T5 in `BlockedQuestions.lean`, one T3 in `ModularTime.lean`, one T5 in `XYZTripotents.lean`) — and the word `admit` in code **zero times**. None of the Version 3 modules introduces either; every other occurrence of either word in the sources is prose, mostly statements that a file contains none. The corpus total of §0 — exactly seven, all pre-existing and labeled — is unchanged.

Axiom footprint (increments). `#print axioms` spot-checks on the Version 3 headline results return exactly `[propext, Classical.choice, Quot.sound]` — no `sorryAx`, no new axioms: nine named headlines of Pitch 2 Campaign A and eight of Campaign B; `hamInt_gap_persistence`, `densityInt_not_dgamma`, and `intOp_isSymmetric` of Pitch 3 Campaign 1; ten named headlines of Schrödinger dynamics; six named headlines of Pitch 3 Campaign 2; six named headlines of the boost discharge (`boostInvariance_window_proved` among them); eight named headlines of the bridge exhibit; eight of Stage 0; the five scoped headlines of Stage 1a (`gibbsExpect_eq_integral`, `elitzur_single_link` among them); six of Stage 1b-B; eleven of Stage 1b-C; and eleven of Stage 1b-D (`transferOp_inner`, `transferOp_isSelfAdjoint`, `transferOp_zero_apply` among them).

Corrections to Version 2 (in the open)

Version 2 is a dated snapshot and is not retro-edited. Four of its statements are overtaken by the work Version 3 reports, and the recount above corrects the running chain built since Version 2. They are listed here, and amended in place in the sections named:

1. **§16, "No infinite-mode or continuum Γ" — FALSE since 2026-10-16.** Pitch 2 (§10A) verifies Γ at countably many modes for the free theory. The line is amended in §16; the continuum clause of it stands.
2. **§16 and the Version 2 abstract, "no interacting theory of either statistics is constructed" — overtaken by Pitch 3 (§10B).** Interacting Hamiltonians at finite modes are constructed for both statistics. What stands: no interacting continuum theory, no gauge-coupled interaction, no scattering. The lines are amended in the abstract and §16.
3. **§7's mass-shell track and §14 item 3 — the priced boost lemma is discharged.** The one-dimensional Jacobian identity is proved and the window closed whole (§7 amendment); §14 item 3 is marked discharged.
4. **§14 item 2 — the infinite-mode limit is discharged in the free theory** by Pitch 2 (§10A); the item is amended to its priced remainder (the spectral gap statement; rectangular functoriality at Task 3b scope).
5. **The counts stand as published.** Under the strict line-initial rule retained for Version 3, the Version 2 program total recounts to exactly its published 1,648, and the Rung 9 count is 50 as printed. What the recount corrects is the running chain assembled since (see the addendum): two landing dossiers counted attribute-prefixed declarations; the strict full-program total is 2,417.

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1. Rung 1 — The finite spectral triple (T1) — STANDS

The finite factor is a concrete matrix spectral triple; its classification and constraint census are machine-checked.

- **Classification and census.** The generator family `smGen : Fin 12` gives 144 pairs; the order-zero condition is checked machine-side over 147,456 complex

components (`order_zero_comm`). Three-generation inheritance is proved in `ThreeGen.lean` (20 declarations), including `smGen3_order_zero` and the Kronecker-factorization identities. The 576-pair order-one census is a T4 Python scan, not a Lean proof.

- **Modular dynamics.** `thermal_flux_vanishes`: the modular flux vanishes identically on Gibbs states for every inverse temperature β. `exists_activeDriver`: a non-equilibrium driver producing nonzero flux exists. `modularFlux_selfAdjoint`: the flux φ = i[K, D] is Hermitian. Gauge covariance and trace gauge-invariance of the flux are proved.
- **Ultimate reduction.** `UltimateReduction.lean` proves 6 T1 theorems, including `dynamicallyFrozen_of_noHeat_noGauge` and `equilibriumKillsFlux`. The same file records finite-level kills: micro-vortices (no vorticity operator exists in the finite triple) and the vortices→matter→gravity chain (needs continuum data).
- **Connes–Kreimer Hopf algebra.** `Renormalization.lean` (60 declarations) formalizes the Hopf algebra of rooted trees: coproduct, B_+ grafting, counity, coassociativity. The record on the antipode is two-sided and both sides are kept: the as-stated antipode axiom was **refuted** (`antipodeAxiom_false`), and the corrected evaluation version was **proved** (`antipodeAxiomEval_proved`, T5→T1). The refuted statement is not laundered into the proved one.

2. Rung 2 — The almost-commutative product (T1) — STANDS

`ProductTriple.lean` (63 declarations) builds the product of an axiomatized manifold factor with the finite triple.

- **Product Dirac identity.** `Dprod_sq`: $D_{prod}^2 = D_M^2 \otimes 1 + 1 \otimes D_F^2$. The formerly axiomatized `product_sum_of_squares` was promoted to a proved T1 — upgraded, not resurrected.
- **Twist derived, not assumed.** The product twist is the flip on the doubled product algebra (`flipProd_involutive`, `flipProd_CM_regular`); the candidate $\rho_M \otimes \rho_F$ product twist was killed (it does not typecheck).
- **Fluctuation decomposition.** `prod_commutator_decomp` splits inner fluctuations into manifold and finite parts.
- **KO data.** `koProd_signs`: KO signs $\langle -1, 1, -1\rangle$ (`koDim_add`: $(4 + 6) \% 8 = 2$).
- **Causality re-entry.** `CausalProduct.lean`: the finite-level Span-5 causality obstruction does not carry to the product (`trace_collapse_hypothesis_fails`, `product_not_forced_central` — T1 consequences of the T2 product axioms, footprint labeled in §0). Genuine causal order is pinned at its exact price: global hyperbolicity.
- **The 6-dimensional caveat.** The surviving finite represented gauge algebra is six real dimensions: Option-A kills the represented $M_3(\mathbb{C})$ summand. The full Standard Model gauge group is not claimed from the finite factor. This caveat stands and is revisited precisely in §11.

3. Rung 3 — Lorentzian-structure pins (T2+T1, one partial) — STANDS

- **Antilinear J — promoted (T2+T1).** The obstruction (bare ℂ-modules; J not even statable) was removed by explicit enrichment of `TripleData` (sesquilinear form; positivity, nondegeneracy, completeness explicitly not assumed; antilinear `Janti` with KO axioms). Product theorems proved T1: `Jprod_sq` ($J^2 = -1$), `Jprod_D` ($JD = DJ$), `Jprod_Γ`.
- **Krein inner product — promoted (T2+T1).** Fundamental symmetry η with $\eta^2 = 1$ and form-self-adjointness proved (`ηprod_sq`, `ηprod_selfadj`). No adjoint operator is constructed and no spacetime signature is claimed: η is an abstract fundamental symmetry.
- **Twisted first-order condition — PARTIAL.** The opposite representation is built in the `conjJ`-shadow of $a^\circ = Ja^*J^{-1}$ (multiplication reverses, T1); the diagonal reduction is proved (`prodTwistedFOC_diag_reduce`). The exact remaining obstruction is priced: no `Star` instance on `Module.End` over the tensor module in this setting, and factor-level order-one is absent from the abstract `TripleData`.
- **Continuum scoping — priced roadmap, no limit claimed.** "Finite triple converges" language is killed with reason; the continuum is scoped per-axiom as future work.
- **Promotion honesty.** The Rung 3 promotions are T2-axiomatization plus T1 consequences — not derivation from analysis. This paper repeats that qualification wherever the promotions are cited.

4. Rung 4 — The analytic house (T1 + one T2 bridge) — STANDS

Directive: "Build the house and use the gaps as windows."

- **Bounded-triple instance for the concrete Pauli model.** `pauliBounded : BoundedAnalyticTriple` forms. Its mathematical content is proved T1 underneath (`pauliD_sub_det`, `pauliD_resolvent`, `pauliD_resolvent_compact` — finite-dimensionality is the whole proof of compactness); one model-specific T2 axiom (`pauli_compactResolvent`) bridges the deliberately opaque `HasCompactResolvent` predicate. The window is eliminable only by redefining that predicate transparently — a design decision, priced in the dossier. The model stays analytically trivial (scalar representation, vanishing commutators); it discharges analytic debt, not gauge content.
- **Hilbert completion and bounded *-representation (T1).** Completion of the model is trivial (`CompleteSpace (EuclideanSpace ℂ (Fin 2))`); the bounded *-representation of the Pauli algebra is proved with the true adjoint star, transported from matrix conjugate-transpose (`repScal_is_star_rep`). The incorrect `star := conjJ` construction was explicitly refused earlier in the program and stays refused.

- **Smeared field operators (14 T1).** Creator-half canonical (anti)commutation relations for smeared fermionic and bosonic fields, smeared Pauli exclusion, and one-point vanishing (`FieldOperators.lean`).
- **Wightman targets framed (T5, unproved by design).** All six axioms are stated as structured propositions with exact hypotheses, conclusions, and per-axiom prices (`WightmanTargets.lean`). Stating them is not proving them; Rung 5 takes the verdicts.

5. Rung 5 — The Wightman ascent on a one-mode model — STANDS

On a one-mode fermionic Fock model ($\wedge(\mathbb{C}) \cong \mathbb{C}^2$, Majorana field $\psi(c) = c \bullet (a^\dagger + a)$), all six targets received T1 verdicts. These are verdicts about the toy model and about the framings — not about 4D QFT.

- **W1 (fields exist): PROVED** (`w1_majorana`).
- **W2 (covariance): PROVED for $\mathbb{Z}_2$ fermion parity** (`w2_parity`). This is explicitly not Poincaré covariance; the group is `ZMod 2` acting by parity.
- **W3 (spectrum condition): PROVED for the trivial translation representation** (`w3_trivial`) — a frame-consistency check (spectrum $\{0\} \subseteq$ forward light cone), not Stone's theorem.
- **W4 (locality): PROVED** at $s = +1$ (`w4_majorana`); the one-mode model has no spacetime, so the spacelike hypothesis is vacuous.
- **W5 (vacuum cyclicity): REFUTED as framed; corrected version PROVED.** Monomial-range density is false even for a free fermion (`w5_framed_false`); the linear-span density the literature means is true (`w5_span_true`). The Rung 4 framing was strictly stronger than the literature; the refutation is a framing correction recorded as a T1 result.
- **W6 (non-triviality): bosonic form proved but fermion-blind; corrected version PROVED.** The bosonic-form criterion gives $-2 \neq 0$ for the free Majorana field — a documented false positive (`w6_majorana`); the corrected Koszul-sign Wick criterion vanishes as it should (`w6_fermionic_wick`).

Rung 5 files: `OneModeFermion.lean` (37 declarations) and `Rung5W1–W6` (39 declarations).

6. Rung 6 — Infinite modes — STANDS

- **Schwartz machinery — premise correction, then GO.** The Rung 4/5 framing "Mathlib has no Schwartz space" was wrong: Mathlib ships the full Schwartz machine (Doll 2026). `SchwartzSpace.lean` proves 14 T1 results on genuine $\mathscr{S}(\mathbb{R}^4, \mathbb{C})$ — closure under derivatives and polynomial multiplication, temperate growth, Fourier inversion — and discharges the earlier T2 test-function marker. One genuine gap remained at this rung: no `CompleteSpace` $\mathscr{S}$`(E,F)` instance in Mathlib (discharged at Rung 7, §7).
- **Poincaré — geometric half GO.** `PoincareAction.lean` (11 declarations): the Lorentz group O(1,3) and Poincaré group as `Group` instances; the field action `f ↦`

`f(Λ⁻¹(x−a))` satisfies the group-action laws and is ℂ-linear in the test function (`poincareAct_mul`, `poincareActLin`). The one-particle unitary representation and second quantization stayed priced at this rung.

- **Wick combinatorics — PARTIAL.** `WickSigns.lean` (31 declarations): complete pairings of `Fin (2n)`, the Koszul sign, its independence of pair ordering, and the four-point signs (+1, −1, +1) (`koszulSign_four_point`). The operator Wick identity stayed priced (discharged at Rung 7 in a two-mode model).
- **Stone — analytic core T1.** `continuous_char_lift`: a continuous norm-one character of (ℝ,+) is `exp(i·a·t)` — the analytic core of Stone's theorem. The full finite-dimensional Stone theorem hit Lean engineering friction at this rung (discharged at Rung 7).

7. Rung 7 — The frontier — STANDS

Four of the five outstanding prices were discharged T1; the fifth was narrowed. This rung also marks where formalized analysis currently stops.

- **Schwartz completeness — T1.** `schwartzComplete_discharge` (`SchwartzComplete.lean`, 415 lines): $\mathscr{S}(\mathbb{R}^n, \mathbb{C})$ is Fréchet-complete — the genuine Mathlib gap from Rung 6, closed by a direct proof (countable uniformity basis → uniform convergence of all iterated derivatives → decay bounds pass to the limit).
- **Operator Wick identity — T1, with a correction.** In a concrete two-mode fermionic Fock model, `wick_four_point_operator`: $\omega(\psi_0\psi_1\psi_2\psi_3) = G_{01}G_{23} - G_{02}G_{13} + G_{03}G_{12}$, with all Wick hypotheses proved (CAR, Hilbert adjoints, quasi-free move), and the Koszul signs discharged against Rung 6. Correction recorded in-file: Rung 6's priced hypothesis pair $(\{\psi_i,\psi_j\} = 2G_{ij}$ together with $\omega(\psi_i\psi_j) = G_{ij})$ is jointly false in the model; the identity holds with the full non-symmetric G. The n-point expansion remains priced (T5).
- **Stone / joint spectral ONB — T1.** `jointSpectral_orthonormalBasis` (commuting symmetric family ⇒ orthonormal basis of joint eigenvectors) and the full finite-dimensional Stone theorem `stone_finiteDim` (strongly continuous unitary one-parameter group ⇒ skew-adjoint generator diagonal in that basis, `(U t)(b k) = exp(i·a k·t) • b k`).
- **Joint functional calculus — T1.** For commuting bounded symmetric operators, `SpectralFrontier.lean` proves a joint functional calculus stronger than Mathlib's single-operator continuous functional calculus, with the spectral mapping theorem `jointFC_spectralMapping`.
- **Mass-shell track — PARTIAL.** The invariant measure $d^3p/2E_p$ is constructed as a density-weighted pushforward; its integral formula and rotation invariance are proved, and the restricted (proper orthochronous) Lorentz group is formed as a subgroup. Boost invariance is priced at exactly one missing lemma: the one-dimensional Jacobian identity $|\varphi'| = E'/E$ for the x^1-boost. The one-particle representation law and unitarity-from-invariance are proved at function level; strong continuity and the linear-isometry packaging remain priced. Second

quantization was blocked at this rung by the Fock inner-product gap — the subject of Rung 8.

- **Amendment (2026-10-17) — the boost window is DISCHARGED WHOLE.** The one missing lemma priced above is proved, and the cap the pricing flagged as likely is proved with it. `MassShellBoost.lean` (1,629 lines; 80 declarations, 15 definitions; targeted build green, 3,463 jobs; six headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`). The x^1-boost is built as a concrete element of the restricted Lorentz group (`boostRestricted`: Minkowski-form preserving, determinant 1, orthochronous); the shell parametrization is intertwined with the boosted-energy identity $E' = \gamma(E - \beta u_1)$; the priced Jacobian identity is proved exactly as scoped — $|\varphi'(u_1)| = E_{\varphi(u_1)}/E_u$, with φ strictly monotone and surjective (`boostSlice_jacobian`, `boostSlice_abs_deriv`) — and the density cancellation $(E'/E)\cdot(2E')^{-1} = (2E)^{-1}$ assembles into `boost_preserves_massShellMeasure`: the boost's shell action is measure-preserving for the mass-shell measure. The cap: **every** restricted Lorentz transformation factors as (spatial lift)$^{-1}$ ∘ (x^1-boost) ∘ (spatial lift) — the spatial factor built from a Householder reflection, with the block lemma that a Lorentz map fixing the time direction is a spatial lift — so `boostInvariance_window_proved` holds for every Λ: **the free one-particle mass-shell measure is invariant under the full restricted Lorentz group.** Scope, exactly: the free, one-particle measure. This rung's other priced items are untouched by the discharge: **strong continuity** of the one-particle representation remains priced (repriced by the 2026-10-17 parked-fleet gauntlet: the translation part is a discharge-ready S — the file itself names the dominated-convergence lemma and the dominator $4\|\psi\|^2$; the Lorentz part is S–M, with this discharge's Jacobian identity supplying the domination), and **Fock packaging** remains priced at M (a concrete orthonormal-basis port; the abstract SymmetricAlgebra route stays killed, §15).

8. Rung 8 — The Fock inner product — STANDS (fermionic) / PARTIAL (bosonic, closed at §8B)

- **Fermionic — STANDS.** `FermionicFock.lean` (43 theorems, 1,064 lines): the Fock space as ℓ^2 over occupation subsets, with Pauli exclusion automatic from the subset guards. Proved T1: creation/annihilation adjoints (`ann_adjoint`, `adag_adjoint` — the fermionic discharge of the W0 inner-product gap), the canonical anticommutation relations, sector orthogonality, the number operator with eigenvalues `S.card`, and vacuum cyclicity. **Fermionic second quantization is unblocked.** Priced windows, stated in-file: unitary equivalence with exterior-power subspaces, the infinite-mode limit, and the general-n Gram determinant (base case and reassembly proved; inductive step priced).
- **Bosonic — PARTIAL at this rung; closed at §8B.** `BosonicFock.lean` (25 theorems): ℓ^2 over finitely supported occupation configurations, with creation/annihilation as linear maps on a dense domain (the honest standard formulation for unbounded operators). The inner product, orthonormal basis, explicit basis actions, and the basis-level adjoint relation (`adjoint_basis` — the computational

core of W0) are proved. The final linearity assembly of the full adjoint relation stalled on Mathlib API friction (submodule inner-product coercion handling) and is priced as engineering, not mathematics; the induction principle needed for the assembly is proved. CCR, number operator, and sector results follow from the basis actions by standard computations not yet formalized. All of these were discharged in full at Rung 8B (§8B), with no weakened statements.

8B. Rung 8B — Bosonic assembly (T1) — STANDS; the Rung 8 bosonic partial is closed

`BosonicAssembly.lean` (520 lines; 31 theorems and 6 definitions, zero sorrys; imports `BosonicFock.lean`, which is unmodified) performs exactly the assembly Rung 8 priced. All results are T1, on the dense finite-particle domain `Dom d` over a finite mode set `Fin d` — the free bosonic theory, stated with the same honesty labels as §8.

- **Full adjoint relation.** `adjoint_relation`: for all `x y : Dom d`, `⟪ann i x, y⟫ = ⟪x, adag i y⟫`, assembled from Rung 8's `adjoint_basis` by its own two-variable induction principle; the Rung 8 API friction yielded to an explicit motive. Companion form `adjoint_relation_adag`.
- **CCR on `Dom d`.** `ccr_mixed` — the mixed commutator `(ann i) ∘ₗ (adag j) − (adag j) ∘ₗ (ann i)` equals `1` if `i = j` and `0` otherwise, as linear maps on `Dom d` — together with `ccr_ann_ann`, `ccr_adag_adag` (annihilators commute; creators commute) and the pointwise form `ccr_mixed_apply`. Proved first on coefficient space and transported through the representation, not assembled by fiat.
- **Number operator.** `numberOp = ∑ i, (adag i) ∘ₗ (ann i)`, with `numberOp_basis`: the basis eigenvalue is the total occupation count, and `numberOp_vacuum`: the vacuum eigenvalue is `0`.
- **Vacuum and sectors.** `ann_vacuum` (annihilation kills the vacuum in `Dom d`); `sector_orthogonal`: sectors of distinct total occupation are orthogonal.
- **Vacuum cyclicity (stretch target, landed).** `cycSub_eq_top`: the submodule of `Dom d` generated from the vacuum by finitely many creation operators is everything.

Priced remainder, unchanged and not attempted here: bosonic Γ(U) (the subject of §10, Task 3); the infinite-mode limit; unitary equivalence with symmetric tensor-power sectors and the Gram/permanent formula (blocked upstream on n-fold tensor inner products); the continuum one-particle space and the mass-shell representation (the §9 Task 4 frontier).

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9. Rung 9 — Fermionic second quantization (T1) — STANDS (Tasks 1–3) / PARTIAL (Task 4)

All results in this rung are T1: 50 theorems and 13 definitions across two files, zero sorrys, with axiom spot-checks returning exactly `[propext, Classical.choice, Quot.sound]` for every headline theorem. Labels that apply to everything below: **free**

theory, finite modes. This rung is not a continuum QFT, not an interacting theory, and not the Yang–Mills mass gap; the one gap in it is derived, and it is labeled exactly where it lives.

- **Task 1 — the Γ lift: STANDS.** `SecondQuantGamma.lean` (992 lines, 32 theorems, 10 definitions). For a one-particle matrix `M` on the mode space `Fin d`, `Γ(M)` is constructed on the Rung 8 fermionic Fock space by the compound-matrix (minor) formula. Proved: `gamma_one` (`Γ(1) = 1`); `gamma_mul` (`Γ(M * N) = Γ(M) * Γ(N)`); `gamma_adjoint` (`Γ(M*) = Γ(M)*`, in inner-product form); `gamma_vacuum` (`Γ(M)` fixes the vacuum, for every `M`); `gamma_sector` (`Γ(M)` preserves every n-particle sector); `gamma_inner` with `gamma_comp_self` and `gamma_self_comp` — for unitary `M`, `Γ(M)` is unitary as an operator identity; and `gamma_diagonal_basis` — `Γ(diag v)` multiplies a basis vector by the product of the selected phases. The supporting determinant core is proved in the same file, from scratch where Mathlib lacks it: `minorDet` and its calculus, the expansion `det_mul_expansion`, and the rectangular Cauchy–Binet formula `cauchy_binet_rect` (not in Mathlib at this toolchain), yielding the compound-matrix identity `minorDet_mul`.
- **Task 2 — the dΓ lift: STANDS.** `SecondQuantDGamma.lean` (329 lines, 18 theorems, 3 definitions). `dΓ(M) = ∑ i, ∑ j, M i j • (a†ᵢ ∘ aⱼ)` as a finite sum of creation–annihilation composites. Proved: `dgamma_adjoint` (the adjoint of `dΓ(M)` is `dΓ(Mᴴ)`); `dgamma_one` (`dΓ(1)` is the Rung 8 number operator); sector preservation; `dgamma_basis_diag` — on a basis vector, `dΓ(diag λ) e_S = (∑ j ∈ S, λ j) • e_S`; and `dgamma_vacuum`.
- **Task 3 — the free Hamiltonian gap, DERIVED: STANDS.** For a diagonal one-particle Hamiltonian with eigenvalues `λ j ≥ 0`, set `freeHamiltonian λ = dΓ(diag λ)`. Proved: the vacuum is an eigenvector with eigenvalue `0` (`freeHamiltonian_vacuum`); every basis vector is an eigenvector with eigenvalue the sum of its occupied one-particle energies (`freeHamiltonian_basis`); self-adjointness; the expectation formula and its nonnegativity; and the derived gap itself — `freeHamiltonian_eigenvalue_gap`: if every one-particle eigenvalue is `0` or `≥ Δ`, then every non-vacuum eigenvalue of `H` is `≥ Δ`, with `free_gap_dichotomy` giving the eigenvalue dichotomy `0` or `≥ Δ`. **Label, precisely:** the one-particle gap `Δ` is an input, but the Fock-space gap is computed from the operator — as the sum of occupied one-particle energies — not assumed as a structure field and not inserted by definition. It is a statement about the **free** theory at **finite modes**. It is not the interacting Yang–Mills mass gap, and nothing in this rung touches the continuum theory.
- **Task 4 — quantum Poincaré implementation at finite modes: PARTIAL.** What lifts is proved, in form: `gamma_rep` — any one-particle unitary representation `ρ : G → Matrix` of any group `G` lifts through `Γ` to a unitary representation of `G` on Fock space (homomorphism, identity, vacuum fixed, sectors preserved, inner product preserved), discharging the form of Rung 6's second- quantization window at finite modes for any group whose action preserves the finite mode set. A concrete instance is implemented: `lattice_translation_rep`, the lattice translation group acting by diagonal phases, whose lift multiplies a basis vector by exactly the product of its occupied modes' phases — the finite-mode shadow of momentum additivity. What does

not lift is priced exactly: (1) the mass-shell one-particle representation (Rung 6's window stays T5; no Lorentz-invariant hyperboloid measure or massive one-particle Hilbert space exists in Mathlib to restrict to finitely many modes, and this rung does not construct one); (2) boosts and continuum rotations (a boost mixes any finite momentum set into infinitely many modes; no finite mode set is preserved, so there is nothing for `Γ` to lift — the liftable symmetry at finite modes is exactly the mode-set-preserving subgroup); (3) the infinite-mode / continuum limit (needs the infinite-mode Fock construction, priced since Rung 8); (4) interaction (`Γ` and `dΓ` are the free lifts; no interacting Hamiltonian, gauge field, or Yang–Mills dynamics appears in this rung).

The free fermionic theory at finite modes is now complete as a machine-checked package: Fock space (§8) + `Γ` + `dΓ` + free Hamiltonian with derived gap + a unitary symmetry representation in lifted form.

10. Rung 10 — Bosonic second quantization (T1) — STANDS (Tasks 1–3)

Two new modules (31 theorems and 13 definitions, zero sorrys; `BosonicFock.lean` and `BosonicAssembly.lean` unmodified). Labels that apply to everything below: **finite modes** (`Fin d`), operators on the **dense domain** `Dom d` (finite-particle vectors), **free bosonic theory**. This is not the interacting Yang–Mills theory and not a Millennium claim. The Task 2 gap is derived for the free Hamiltonian at finite modes — exactly the bosonic analog of §9 Task 3, with the same scope.

- **Task 1 — the bosonic dΓ lift: STANDS.** `BosonicSecondQuant.lean` (531 lines, 25 theorems, 5 definitions). `dgammaBos d M = ∑ i, ∑ j, (M i j) • ((adag d i) ∘ₗ (ann d j))` on `Dom d`, mirroring §9's fermionic `dgamma` with the Rung 8B domain adjoint relations in place of finite-dimensional adjoints. Proved: `dgammaBos_adjoint` (`⟪dΓ(M) x, y⟫ = ⟪x, dΓ(Mᴴ) y⟫` on `Dom d`); `dgammaBos_one` (`dΓ(1) = numberOp`, connecting to §8B's number operator); `dgammaBos_vacuum`; `dgammaBos_basis_diag` (`dΓ(diag λ) e_c = (∑ i, λ i * (c i : ℂ)) • e_c`); sector preservation (`dgammaBos_mem_domSector`), with `domSector_coe` coercing the domain sectors into §8B's Fock sectors.
- **Task 2 — the bosonic free Hamiltonian gap, DERIVED: STANDS.** For `freeHamiltonianBos d lam = dΓ(diag λ)` with `λ i : ℝ`, proved: every basis vector is an eigenvector with eigenvalue `∑ i, λ i * (c i)` (`freeHamiltonianBos_basis`, with the coefficient form `freeHamiltonianBos_coord`); the vacuum eigenvalue is `0`; self-adjointness on `Dom d`; the expectation formula (`freeHamiltonianBos_expectation`) and its nonnegativity for `λ ≥ 0`; and the derived gap — `free_gap_dichotomy_bos` and `freeHamiltonianBos_eigenvalue_gap`: every eigenvalue of `H` on a nonzero vector of `Dom d` is a basis eigenvalue, hence `0` or `≥ Δ` whenever each one-particle eigenvalue is `0` or `≥ Δ`. As in §9, the one-particle gap `Δ` is an input and the Fock-space gap is computed from the operator; the statement is about the **free** theory at **finite modes**.

- **Task 3 — bosonic Γ(U): PARTIAL at the rung's first landing; CLOSED by the Γ functor campaign (2026-10-16, later the same day).** `BosonicGamma.lean` (215 lines, 6 theorems, 8 definitions). The fermionic Γ of §9 used determinant minors over subsets; bosons need **permanent** minors over occupation configurations, repetitions allowed. The lift is defined (`configsOfTot`, `configEnum`, `perMinor`, `normFactor`, `gammaKernel`, `gammaC`, `gammaBos`) and what the definition gives directly is proved: the kernel formula `gammaBos_apply_basis` (`Γ(M) e_c` is the sum, over configurations `c'` of the same total count, of `gammaKernel M c' c • e_{c'}`); `perMinor_of_tot_ne` (minors vanish across different total counts); `gammaBos_sector` (`Γ(M)` preserves the total-occupation sector); and `gammaBos_vacuum` (`Γ(M)` fixes the vacuum). **Priced remainder — the two obstructions, named exactly:** (1) `Γ(1) = 1` and the diagonal action `Γ(diag λ) e_c = (∏ i, λᵢ^{c i}) • e_c` both reduce, for this kernel, to the **fiber-permanent counting lemma**: the permanent of the fiber-indicator matrix (entries `1` where the enumerated positions of `c` agree, else `0`) equals `∏ i, (c i)!` — a finite but genuine counting argument about `Matrix.permanent` over repeated indices, not in Mathlib at this toolchain. (2) **Multiplicativity** `Γ(M)Γ(N) = Γ(MN)` and **unitarity** need a **permanent Cauchy–Binet identity for repeated indices**: expanding `per (M N)` over intermediate index functions and grouping them by their multiset image, with the `1/∏ c'!` weights matching the kernel normalization exactly — the bosonic analog of §9's rectangular Cauchy–Binet campaign, a separate campaign of comparable size. Until (1) lands, even `Γ(1) = 1` is unproved for this definition — so bosonic Γ is recorded as a candidate lift with its kernel formula, vacuum, and sector preservation proved, not as a verified functor. No statement was weakened to get here. **Closure (2026-10-16):** the functor campaign (`BosonicGammaFunctor.lean`, 1,161 lines, 42 theorems, 8 definitions, zero sorrys; targeted build green, 2,455 jobs; headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`) proved both obstructions with their consequences: the fiber-permanent counting lemma (`permanent_fiberIndicator_configEnum`), `Γ(1) = 1` (`gammaBos_one`), the diagonal action (`gammaBos_diagonal_basis`), the permanent Cauchy–Binet identity for repeated indices (`perMinor_mul`, with the multiplicity-weight bookkeeping proved), multiplicativity (`gammaBos_mul`), and the unitarity package (`gammaBos_adjoint`, `gammaBos_comp_conjTranspose`, `gammaBos_conjTranspose_comp`, `gammaBos_isometry`). **Task 3 is closed; bosonic Γ is a verified functor on finite modes, and Rung 10 stands whole.** Full record: `whitepaper/gamma-functor-campaign-2026-10-16.md`.

Priced remainder beyond this rung: the infinite-mode limit (no longer `Fin d`); the symmetric tensor-power unitary equivalence / permanent formula in Mathlib's tensor API (still blocked upstream, as priced at §8); and the standing frontier — continuum limit, interacting theories, Yang–Mills — which nothing in this rung touches.

10A. Pitch 2 — The free theory at countably many modes (T1) — STANDS WHOLE

The priced build order for the infinite-mode limit that §10 left open — scoped in `whitepaper/pitch-2-infinite-mode-scoping-DRAFT-2026-10-16.md` and executed in two campaigns (2026-10-16) — landed in full. The landing removes exactly one scope word from §§9–10: "finite modes" becomes "countably many modes." Everything else about the labels is unchanged: **free theory**, states on the **finite-particle domain**, and the gap in the sense stated below. Five new modules, 137 theorem/lemma declarations and 57 definitions, zero sorrys; headline axiom checks (nine in Campaign A, eight in Campaign B) exactly `[propext, Classical.choice, Quot.sound]`.

- **Campaign A — the tower (57 declarations).** `FockLimit.lean` builds the shared machine Mathlib lacks: `onbEmb`, a linear-isometry constructor from a source Hilbert basis and a target orthonormal family, and density of finite-support vectors in the constant-$\mathbb{C}$ `lp` spaces. `FermionicLimit.lean` and `BosonicLimit.lean` build the limit spaces — ℓ^2 over configurations on $\mathbb{N}$ (Finset $\mathbb{N}$ for fermions, $\mathbb{N} \to_0 \mathbb{N}$ for bosons) — with orthonormal configuration bases via `lp.single`, vacua, finite-particle domains and sectors; configuration embeddings between all finite levels and into the limit, with the cocycle proved in the **general** coherence form (all $d \le e \le f$, not merely successive steps) and the factorization that every limit configuration lives at a finite level. (A forced split, disclosed: `FermionicFock` and `BosonicFock` cannot share one Lean environment — both define `ThetLogos.adag`; the limit files inherit the split. The collision is the subject of the priced `adag` repair, §14.)
- **Campaign B — the operators and the gap (80 declarations).** `FermionicLimitOps.lean` and `BosonicLimitOps.lean`. The corpus Γ API is square-only, so a rectangular widening was ported (`minorDetRect`/`perMinorRect`, definitionally the square kernels) and the identifications proved at operator level: `gammaFermRect_pad` (whole-space operator equality, fermions) and `gammaBosRect_pad` (equality of domain maps — the honest bosonic landing, since bosonic Γ preserves domains, not whole spaces). The finite-particle domain of the limit is the union of the finite levels (`fermDomLim_eq_iSup_range`; `DomLim_eq_iSup_map` in domain-map form — the bare-range form is false for bosons, recorded), with density corollaries. Diagonal $d\Gamma$ in the limit for both statistics: basis eigenvalue formulas, finite-level compatibility as operator equalities, and the number operator as the constant-1 case.
- **The gap, Δ-general (Task 6).** For a diagonal one-particle Hamiltonian whose eigenvalues are 0 or $\ge \Delta$ (any $\Delta > 0$), the free Hamiltonian in the limit has vacuum eigenvalue 0 and the eigenvalue dichotomy: every finite-particle eigenvalue is 0 or $\ge \Delta$ (`freeHamiltonianLimFerm_gap_of_orth`, `freeHamiltonianLimBos_gap_of_orth`), with self-adjointness on the domain. **Mandatory clause, carried in both module headers and repeated here verbatim in substance: this is the gap in the eigenvalue-on-finite-particle-domain sense of Rungs 9–10; the spectral statement (the gap as a statement about the full spectrum of the closed operator) is priced,**

unattempted — the scoping's Task 7, a pre-agreed cut, as is Task 3b (rectangular functoriality beyond the padding identifications proved).

What Pitch 2 does not buy (its dossier's honesty line, adopted): not interaction — that is Pitch 3 (§10B), which is a separate construction at finite modes and has no limit-space form; not continuum or Poincaré (§13A); and not any part of any Millennium statement.

10B. Pitch 3 — Interaction at finite modes — STANDS (fermionic, whole) / STANDS except one priced clause (bosonic)

The corpus's first interacting Hamiltonians. Both campaigns are finite-mode matter self-interactions — quartic Wick-ordered density-type terms added to the free Hamiltonians of §§9–10. No gauge field, no continuum, no Poincaré, and not the interacting Yang–Mills mass gap: that clause is carried in both module headers and applies to everything below. Scoping: `whitepaper/pitch-3-interaction-scoping-DRAFT-2026-10-16.md` (§6 named the landing shape and its failure consequence in advance; the verdict below applies §6 exactly as written).

- **Campaign 1 — fermionic interaction: STANDS WHOLE.** `FermionicInteraction.lean` (1,303 lines; 58 declarations, 15 definitions; build green, 3,297 jobs; headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`). The general interacting Hamiltonian `hamInt lam W g = freeHamiltonian + g • intOp W` is symmetric for every coupling tensor W, with g = 0 recovering the free Hamiltonian; the quartic `intOp W` is built on the corpus's first operator-norm bounds for the one-mode creators and annihilators, with every matrix coefficient and row sum of `intOp W` bounded by `intBound W = 2·∑‖W i j k l‖`. **Gap persistence (headline):** if `2·|g|·intBound W ≤ Δ/4`, there is a ground eigenvalue E_0 with `|E₀| ≤ |g|·intBound W`, and every eigenvalue is either in the ground cluster or `≥ Δ/2` — the free gap persists at size ≥ Δ/2 above the ground cluster, in the eigenvalue sense, at finite modes (`hamInt_gap_persistence`). The interaction is proved interacting: `densityInt_not_dgamma` — the density interaction equals `dΓ(M)` for no one-particle matrix M. Recorded in the module, per the house pattern: a scoped exact-gap statement under the weak dichotomy was **false** (two zero modes with U a b > 0 give an eigenvalue g·U a b ∈ (0, Δ)); the counterexample is in the theorem's doc-comment, the exact gap is proved under the strengthened no-zero-mode hypothesis, and the true weaker statements are proved instead.
- **Campaign 2 — bosonic interaction: STANDS except essential self-adjointness.** `BosonicInteraction.lean` (1,956 lines; 69 declarations, 21 definitions; targeted build green; headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`). **B1 (Wick calculus):** monomial calculus on the finite-particle domain `Dom d` with the √-coefficient basis formula and number-relative bounds (`intBoundBos W = 2 · couplingMassBos W`); the symmetrized quartic `intOpBos W` is symmetric on `Dom d` for every W (`intOpBos_isSymmetric`). **B2**

(self-adjointness, in three scoped senses): symmetry of `hamIntBos lam W g` on the dense domain; sectorwise self-adjointness with full diagonalization (on every finite particle sector the operator is symmetric with an orthonormal eigenbasis and real eigenvalues — `hamIntBosSector_spectral`); and LinearPMap formal self-adjointness, `hamPMap ≤ hamPMap†` (`hamPMap_le_adjoint`). **B3(i) (exact diagonal gap):** under the **strong dichotomy** (`∀ i, Δ ≤ lam i` — no zero modes), every eigenvalue of the diagonal bosonic Hamiltonian is exactly 0 or ≥ Δ, uniformly over all sectors (`hamiltonianDensBos_gap_exact`). The weak-dichotomy form is **false**, by the same two-zero-mode counterexample shape as Campaign 1, recorded in-doc; only its dominated form is proved. **B3(ii) (windowed persistence):** for an eigenvector in sector n, under the threshold `2 · (‖g‖ · intBoundBos W · (n+2)²) ≤ Δ/4`, the eigenvalue is within `‖g‖ · intBoundBos W · (n+2)²` of 0 or ≥ Δ/2 — the window is in the statement (`hamIntBos_gap_persistence_windowed`). The density interaction is again proved not to be a dΓ lift (`densityIntBos_not_dgamma`) — for both statistics now.

- **Disproved in-module (a result, recorded as one).** Uniform gap persistence for general bosonic quartic coupling is **false**: `densHamEigBos_unbounded_below` — attractive pair coupling (U a b = −1, g = 1) gives the eigenvalue (lam a + lam b)·n − n² at the (n, n) configuration, below every bound for large n. This proved collapse is the reason B3(ii) is windowed; it is also entered in the kill ledger (§15).
- **The one open clause — essential self-adjointness: PRICED.** What is not proved is the closure equality `hamPMap = hamPMap†` as partial linear maps. The obstruction, with one correction to the campaign dossier's account: the dossier stated that Mathlib has no LinearPMap closure operator; the parked-fleet gauntlet's audit (2026-10-17) found that stale — pinned Mathlib does define `LinearPMap.closure` (`Mathlib/Topology/Algebra/Module/LinearPMap.lean`), with `le_closure` and `IsClosable.closure_isClosed`, so the target is statable sharply as `hamPMap.closure = hamPMap†`. What remains absent is the analytic input: no Kato–Rellich theorem and no Nelson commutator theorem in the pinned Mathlib, and the closure's domain is an ℓ²-growth condition across infinitely many sectors. Two proof routes are named: R1, a Faris–Lavine/Nelson commutator argument with comparison operator (N+1)²; R2 (**recommended**), banded truncation — the quartic moves sectors by Δn ∈ {−4,−2,0,2,4} only, so the core property reduces to one ℓ² tail estimate. Re-priced **M–L** (restating the target: S).

> **Pitch 3 verdict** (the scoping §6's pre-named PARTIAL, narrowed to a single clause): Pitch 3 **stands for fermionic interaction whole**. For bosonic interaction, the Wick calculus with relative bounds, the exact diagonal gap, windowed persistence, sectorwise self-adjointness, and LinearPMap formal self-adjointness all stand; **essential self-adjointness is priced**. The §6 headline sentence ("…both statistics, with self-adjointness proved…") is **not** spendable in unqualified form, and no text in this paper spends it.

Route C of the scoping (interaction in the limit spaces of §10A) stays FRONTIER and is not attempted.

10C. Schrödinger dynamics — the evolution exhibit (T1) — STANDS

`SchrodingerDynamics.lean` (868 lines; 49 declarations, 10 definitions; build green, 3,319 jobs; ten headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`). Finite dimensions only; the module takes no interpretation position (interpretations are T5), and whole-Fock-space/bosonic dynamics is FRONTIER and absent from it.

For any symmetric operator T on a finite-dimensional complex inner-product space, `evolOp hT t` is constructed diagonally — spectral phases in Mathlib's `eigenvectorBasis`. Proved: the group law (`evolOp_add`, with the inverse identities) and `evolOp_zero`; unitarity (`evolOp_isometry`; the linear isometry equivalence `evolUnitary` with inverse `evolOp (−t)`; `evolOp_adjoint`); the explicit solution formula (`evolOp_solution_formula`: coefficients evolve by eigenphases); **the Schrödinger equation as a derivative theorem** (`hasDerivAt_evolOp_apply`: dψ/dt = −i • Tψ) with **uniqueness** (`evolOp_unique`, by a coordinatewise integrating factor — no ODE library needed); eigenspace invariance. The Heisenberg side is built with its group law and algebra-automorphism package, and the Heisenberg equation (dA/dt = i[H, A] in the corpus's sign conventions). **Picture equivalence is a theorem** (`picture_equiv`): expectation values agree between the Schrödinger and Heisenberg pictures. **Headline application:** under the exact hypotheses of §10B's `hamInt_gap_persistence`, the ground cluster and the high (≥ Δ/2) cluster of the interacting fermionic Hamiltonian are subspaces, each invariant under the time evolution, covering the space with the ground cluster nontrivial (`groundCluster_invariant`, `highCluster_invariant`, `hamInt_clusters_cover_and_nontrivial`) — the gap persists not only under coupling but in time.

Two fenced shortfalls, named and priced: `evolOp` is characterized as the exponential by the Schrödinger equation plus uniqueness, but a literal identification `evolOp = NormedSpace.exp (−it•T̂)` was **not** proved (priced **S**); and the Dyson first-order term (interaction-picture, needs interval integrals of the Heisenberg-evolved interaction) was **not** landed (priced **S–M**).

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11. SM-Match — structural match against the Standard Model (3 MATCH + 1 PARTIAL)

"Match" means: the formalized structures coincide with the Standard Model's known algebraic structure — same gauge group, same fermion quantum numbers, same classical Lagrangian shape, same breaking pattern. Masses, couplings, and mixing angles are **inputs** (labeled T2), never derived. This is a structural formalization result, not a new Standard Model and not a derivation of any measured value. 95

theorem/lemma declarations across four files (the program dossier's earlier estimate of "~70" understated the count; the verified figure is 95).

- **Track 1 — Full finite gauge algebra: PARTIAL.** On the algebraic side, the match is proved T1: the unitary group of $\mathbb{C}\oplus\mathbb{H}\oplus M_3(\mathbb{C})$ is U(1)×SU(2)×U(3) (`quat_mem_specialUnitary_iff`: $U(\mathbb{H}) = SU(2)$), and its unimodular subgroup is exactly U(1)×SU(2)×SU(3) (`SMUnimodular_mem_iff`). On the represented side, the Rung 2 caveat is located exactly, not redefined: the kernel of the Option-A representation is characterized (`pi_ker_iff`); $\mathbb{C}$ and $\mathbb{H}$ survive faithfully; $M_3(\mathbb{C})$ survives only locked to the electroweak sector; pure color dies. And the priced gap is proved, not asserted: `pi_u1_eL_trivial` — the represented U(1) acts trivially on the left electron, so **the represented U(1) is not the Standard Model hypercharge U(1)**. A faithful representation of the full SM gauge group on the finite Hilbert space is not achieved; the gap is priced at the pre-Option-A representation or the manifold factor.
- **Track 2 — Fermion content: MATCH.** The eight fermion types carry the Standard Model's (Y, T^3, color) assignments as labeled T2 inputs (hypercharges are not derived; generation triplication is imposed, per `ThreeGen.lean`). Proved T1: electric-charge consistency $Q = T^3 + Y/2$ for all eight types (`charge_consistency`), and consistency of the index↔fermion mapping with the finite-triple embedding. Priced: full SU(3) color is unrepresented in the Option-A representation (Track 1's gap).
- **Track 3 — Classical Lagrangian sectors: MATCH.** 19 declarations: three-factor gauge kinetic term with joint gauge invariance proved (`smGaugeKinetic_gauge_invariant`), the correct −¼ sign, and nonnegativity; the fermion term's fluctuation split (`fermionTerm_split`); the Higgs potential's minima structure (pure algebra); the Yukawa sector structure with an input matrix and no values. Priced: the quantum Lagrangian, the continuum integral, the junk quotient, and all coupling values.
- **Track 4 — Symmetry breaking: MATCH.** 42 declarations. The VEV stabilizer is exactly span(Q) (`higgs_stabilizer_is_Q`) — the precise algebraic statement of $SU(2)_L\times U(1)_Y \to U(1)_{em}$; the VEV minimizes the potential; the photon direction is massless because Q is unbroken (`higgs_photon_direction_massless`); $M_W = gv/2$ and $M_Z = \sqrt{(g^2+g'^2)}v/2$ hold as formal relations among the input parameters (g, g′, v), with no numerical values anywhere.

12. Theories intake, Round 1 — fit verdicts

Standing rule: a theory is admitted only if its core structures map onto the program's formalized objects with real theorems. Refusals are results.

- **KMS / Tomita–Takesaki: FIT at finite level.** 10 T1 theorems in `KMSBridge.lean`, including the finite-dimensional KMS identity for Gibbs states (`gibbs_kms_weight`) and the equivalence of KMS equilibrium with vanishing thermal flux; two previously pinned T5 targets in the modular-flux file are discharged. Priced T5 window: unbounded Tomita–Takesaki theory and type-III factors — Mathlib has no von Neumann algebra theory.

- **Classical Yang–Mills: FIT classically; quantum NO-FIT.** 16 T1 theorems in `ClassicalYangMills.lean`, including the represented curvature from 1-form data with exact gauge covariance (`repCurvature_gauge_covariant`) and gauge invariance of the Tr(F†F) action. The quantum theory — quantization, continuum, mass gap — is recorded NO-FIT and remains quarantined T5.
- **AQFT / Haag–Kastler: per-axiom.** In `HaagKastlerFree.lean` (14 declarations): isotony FIT (T1); locality FIT at creator level and for fermionic even observables (T1), with the honest caveat that the spacelike hypothesis is vacuous at creator level; Poincaré covariance PARTIAL (geometric half T1, implementer priced); vacuum PARTIAL (cyclic vector T1, invariance priced); spectrum condition and irreducibility NO-FIT as theorems in the current setting (priced windows recorded).
- **Spectral action / finite index bridge: FIT at finite level (2026-10-11 verification); build failure found and repaired 2026-10-15 — see the correction below.** On the formalized `smDirac`: Tr(1) = 32; Tr(D_F^2) as the explicit Yukawa-sum formula; the fluctuated trace expansion; vanishing of odd traces; the finite-dimensional Fredholm index and the chiral-compression index (zero). NO-FIT recorded: the continuum heat-kernel expansion and the Atiyah–Singer theorem (both need continuum elliptic analysis). **Correction (2026-10-15):** re-verification for this paper found that `SpectralActionFinite.lean` no longer compiles under the pinned toolchain — four proof failures (`simp` made no progress / failed rewrite at lines 942, 1006, 1024, 1074), all inside the a_2 trace-computation proofs and consistent with Mathlib deprecation drift (`if_pos`/`if_neg` simp behavior). `SpectralActionBridge.lean` fails transitively. The theorem statements are unchanged and no `sorry` was introduced, but until the proofs are repaired this entry's T1 status is **not currently machine-verified** and the FIT verdict is suspended, not standing. Repair is a queued, well-localized job. **Resolution (2026-10-15):** the repair was completed the same day — proof-local fixes only (re-applying the diagonal multiplication reduction inside the affected trace computations); no statement changed; no `sorry`/`admit`/axiom added; the file builds green again and **the FIT verdict stands restored** (theories-intake dossier amended; repair commit `6b4e2f9`).

Round 2 queue (not yet run): BRST/anomalies; the Osterwalder–Schrader reconstruction bridge; noncommutative integration / Dixmier trace.

Note (2026-10-17). Round 2 has still not been run as an intake. Its Osterwalder–Schrader item has meanwhile split (per the parked-fleet gauntlet's ruling): the fit-audit half is killed as redundant — Pitch 4's Stage 1b (§13A) is now the program's OS work, done as construction rather than audit — while the re-pinning half stands: the corpus's `OsterwalderSchrader.lean` header premise ("Mathlib has no tempered distributions") is stale (pinned Mathlib has $\mathscr{S}$ and the Fourier transform; the true absences are the smearing map, the full field φ(f), and OS verification), and the reconstruction dependency audit remains priced S–M. BRST stands priced M; the Dixmier trace item stands priced S–M.

13. The Morley edge-signature gauntlet — PROMISING, with one precise kill

Hypothesis (Jeff): a 4D boundary carrying two 2D edge operators generates the 3+1 split, with the torsion/twist in the D_F projection carried, not assumed away. Steelmanned, then formalized in `EdgeSignature.lean` (30 theorem/lemma declarations; the gauntlet dossier groups the core results into four sections). Build status: verified green by the gauntlet on 2026-10-11; the 2026-10-15 group rebuild timed out without errors, and the file was **re-verified later on 2026-10-15 by a standalone build (exit 0; zero `sorry/admit`; axiom checks on the 13 headline theorems clean)** — see §0. The verdicts below stand as the record.

- **Morley core — T1.** Edge-pair ⟺ fundamental symmetry: from $\eta^2 = 1$, the operators $E\pm = (1\pm\eta)/2$ are idempotent, orthogonal, and complementary (`edgePair_of_symm`); conversely a perfect pair gives $\eta = E_+ - E_-$ with $\eta^2 = 1$ (`symm_of_edgePair`). The "arbitrary input → perfect structure" pattern has real formal content: perfection is forced by $\eta^2 = 1$ alone.
- **Flip twist carried — T1.** The doubled edge decomposition is flip-invariant for (η, η): the Rung 2 twist preserves the perfection.
- **Torsion carried exactly — T1.** $(\eta+T)^2 = 1 + (\eta T + T\eta + T^2)$ (`torsion_defect`); fluctuated edge operators miss idempotence by exactly $(2^{-1})^2 \bullet (\eta T + T\eta + T^2)$ (`edgePlus_torsion_defect`); perfection survives torsion iff $\eta T + T\eta + T^2 = 0$, a Maurer–Cartan-type flatness condition (`torsion_flat_iff`). (The earlier $\mathbb{C}^7$-torsion claim remains killed and is not resurrected.)
- **Strong emergence — KILL, by T1 counterexample.** `signature_not_determined`: (1,1) and (1,−1) on $\mathbb{C}\times\mathbb{C}$ both square to 1, but one has vanishing minus-edge (signature (2,0)) and the other does not (signature (1,1)). $\eta^2 = 1$ does not determine the signature; the 3+1 numbers are extra data. The kill is itself a theorem.
- **Analogy assessment:** real formal content at the algebraic level; the "miracle" grading (maximal symmetry from no symmetry) remains philosophy, labeled NEEDS-WORK as an analogy.

Priced next steps: a D_F-specific torsion map (η built from Dirac data, defect computed under inner fluctuations); the dimensional reading — "two 2D" suggests neutral signature (2,2), and what breaks it to (3,1) is open; integration with the twisted first-order condition; a boundary spectral triple as the formal home of the "4D boundary."

13A. Pitch 4 — The continuum passage, opened on the lattice — Stages 0, 1a, 1b stand as stated; NO CLAIM on the pitch

Pitch 4 is the program's scoped route from its finite constructions toward the continuum Yang–Mills theory. Its scoping (`whitepaper/pitch-4-continuum-scoping-`

DRAFT-2026-10-17.md) is chartered **NO CLAIM**: before every stage below, and after all of them, **parts of the official Clay statement closed: 0**. The scoping's decisions, stated once here: the route is **lattice first** — the analytic route (continuum fields via Minlos-type measure constructions) is priced XL with no landing, blocked at its first move by the absence of nuclear spaces and a Minlos theorem from the pinned Mathlib. The stage map: Stage 0, finite lattice gauge theory; Stage 1a, correlations; Stage 1b, reflection positivity and its consequences; Stage 1c, strong-coupling clustering (priced XL, separately killable); Stage 2, one exact coarse-graining step (M–L; the iterated flow with estimates is XL, Balaban-scale); Stages 3a/3b, OS-reconstruction scaffolding (M / M–L); Stage 3c, uniform estimates along a continuum trajectory — **priced research-scale and open in print**, with one audit rule the scoping states plainly: deriving the continuum gap from a gap assumed uniform along the trajectory is circular as an audit instrument and is forbidden in this program's writing; Stage 4, an interacting Poincaré representation in the continuum — FRONTIER. Two traps are named in advance: Elitzur's theorem (gauge-variant quantities cannot carry expectations — met, and discharged, at Stage 1a) and triviality (the φ^4_4 precedent: a lattice route can end in a free continuum theory; nothing below speaks to that question).

All stages below are finite-lattice T1. The lattice is not the continuum, no limit of any kind is taken or claimed in any of them, and none of them is a step in a Clay proof — they are the first priced bricks of the road the scoping draws.

- **Stage 0 — finite lattice gauge theory: STANDS WHOLE.** `LatticeGauge.lean` (1,656 lines; 68 declarations by the §0 strict method — 72 counting its four `@[simp]` loop lemmas, which the strict rule excludes; 33 definitions and instances; build green, 3,390 jobs). Sites form an abstract finite type Λ with four shift bijections, concretely the torus `Fin 4 → ZMod N`; links are group-valued, `U : Link Λ → G` for a general compact group G; the plaquette and Wilson action are built from a general unitary representation ρ. Proved: **exact gauge invariance of the Wilson action for every gauge transformation** — no gauge fixing anywhere — with invariance of the product-Haar configuration measure in the strong form (`Measure.map (gaugeAction σ g) (configMeasure μG) = configMeasure μG`); Wilson loops gauge-invariant; the normalized Haar probability `haarProb` built in-file from Mathlib's Haar measure, its right- and inversion-invariance proved from uniqueness. The **β = 0 package at general (G, ρ)**: at zero coupling the Gibbs expectation is the product-Haar expectation, the single-plaquette expectation factorizes through the Haar intertwiner P, and `gibbsExpect_zero_plaquette : ⟨W_p⟩₀ = tr P` — which vanishes exactly when ρ has no nonzero fixed vectors (`gibbsExpect_zero_plaquette_eq_zero`), with the circle and full-unitary instances discharged on the torus. Action nonnegativity for $\beta \geq 0$.
- **Stage 1a — correlations and Elitzur: STANDS WHOLE** (pitch-level shape after this stage: PARTIAL-β). `LatticeCorrelations.lean` (1,667 lines; 68 declarations, 13 definitions; build green, 3,391 jobs; the five scoped headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`). The Gibbs measure is constructed as an actual probability measure at **every real β**

(`gibbsMeasure`, with `isProbabilityMeasure_gibbsMeasure`), and Stage 0's ratio-of-integrals expectation is proved equal to its integral (`gibbsExpect_eq_integral`) — the measure Stages 1b–1c consume. The expectation algebra (constants, additivity, scaling, finite sums, $|\langle F\rangle| \le \langle|F|\rangle$, the real form with monotonicity) and the covariance package: symmetry, the centered form, **variance nonnegativity** (`gibbsVar_nonneg`), and **Cauchy–Schwarz** (`gibbsCov_sq_le`, by the elementary discriminant argument). **Elitzur's theorem at general (G, ρ), for every β** (headline): the mechanism theorem `expectMatrix_eq_zero_of_left_covariant` — the expectation matrix of any left-covariant insertion is zero whenever ρ has no nonzero fixed vectors — with corollaries `elitzur_single_link` and `elitzur_open_wilson_line` and the torus instances. The pre-authorized narrowing to instance-level Elitzur was **not** needed. The contrast is proved, not gestured: plaquette expectation matrices are intertwiners, not vanishing subjects (`expectMatrix_plaquette_intertwines`). β = 0 factorization for disjointly supported observables landed **stronger than scoped** — the identities carry no integrability hypotheses at all — and is fenced to β = 0 in the module, its dossier, and here. The Wilson two-point function is defined as an object, with the bound $|\langle W_1 W_2\rangle| \le d^2$ and the connected function defined; **no decay or clustering claim is made at any β** (that is Stage 1c, untouched).

- **Stage 1b-A — reflection structure: STANDS WHOLE.** `LatticeReflection.lean` (1,335 lines; 67 declarations, 16 definitions; build green, 3,392 jobs). The abstract-Λ presentation of Stages 0–1a provably carries no reflection (a grep-verified obstruction recorded by the gauntlet), so the reflection is built on the concrete torus: the **between-sites** reflection in direction 0, `reflSite x = update x 0 (-1 - x 0)` — an involution with the shift calculus $\theta\sigma_0\theta = \sigma_0^{-1}$, $\theta\sigma_i = \sigma_i\theta$ ($i \neq 0$), and **no fixed sites at even N**. The fixed link stratum is **characterized by an if-and-only-if** (`reflLink_eq_self_iff`): the fixed links are exactly the time links based in the two layers adjacent to the two reflection planes — not enumerated, characterized; the pre-named enumeration-only PARTIAL shape was not triggered. The configuration reflection factors as θ = invConfig ∘ permConfig, and **both the product-Haar measure and the Gibbs measure are reflection-invariant at every real β**, in the strong `Measure.map` form (`gibbsMeasure_map_reflConfig`) — stronger than the pre-named Haar-level fallback. **The Wilson action split is proved term by term**: $S = S_+ + S_- + S_0$, with $S_+(\theta U) = S_-(U)$, $S_-(\theta U) = S_+(U)$, and S_0 — the crossing class of temporal plaquettes over the two fixed layers — θ-invariant term by term.
- **Stage 1b-B — reflection positivity: PARTIAL (the pre-named shape γ, extended).** `LatticePositivity.lean` (1,559 lines; 44 declarations, 10 definitions; build green, 3,393 jobs). The observable class is the **strict positive half** `posLinks = halfLinks \ planeLinks`; the exclusion of the reflection-fixed plane links is forced — two finite counterexamples are recorded in the stage dossier (hand computations, not Lean theorems): a plane-link observable over $G = \mathbb{Z}/3$ at β = 0 gives the form value −2, and at fixed stratum variables a U(1) character gives $I_1(\beta)e^{i\gamma}$, not even real. What stands, at general (G, ρ) and **every real β**: the OS form `osForm` (θ on observables = conjugation ∘ the

observable ∘ the configuration reflection), its **conjugate symmetry** (`osForm_conj_symm`, via 1b-A's Gibbs reflection invariance), reality of the diagonal, full sesquilinearity, the bridge to the bare-pullback reading of ⟨θ(F)·F̄⟩ (`osForm_bare_eq`), and the **kernel identity** (`osForm_eq_kernel`): the form is Z^{-1} times a double integral over the two halves, with the crossing action in the kernel. **Positivity is proved at β = 0** (`osForm_zero_nonneg`: the weight is 1, the kernel factors, and the diagonal is W·W̄). The **Gram criterion is proved, β-sign-free** (`osForm_nonneg_of_gram`): if the Gibbs weight in split variables is a finite Gram sum $\sum_i$ conj(Ξᵢ∘mirror)·Ξᵢ∘restrict with bounded measurable features, then the diagonal is $Z^{-1}\cdot\sum_i\|\int f\cdot\Xi_i\|^2 \geq 0$. Positivity at general β is thereby reduced, losslessly, to **one named open clause**: the Gram (character-expansion) factorization of the Wilson weight. Its standard proof needs Schur/Peter–Weyl orthogonality of matrix coefficients on a general compact group, which the pinned Mathlib (v4.35.0-rc3) does not contain — its representation theory covers finite groups and the algebraic theory. **Price: XL, blocked on Mathlib infrastructure.** The positivity theorem itself is classical (Osterwalder–Schrader; Glimm–Jaffe); it is unproved here, **not** killed.

- **Stage 1b-C — the OS Hilbert space: STANDS.** `LatticeOSSpace.lean` (1,065 lines; 42 declarations, 17 definitions; build green, 3,550 jobs; eleven headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`). Built by the **time-slice L² route** (the gauntlet's ruling; the quotient/completion-of-the-form route was declined): `osSpace μG ρ β := Lp ℂ 2 (sliceMeasure μG ρ β)`, where the slice measure is the Gibbs marginal on the time-0 slice — a probability measure at **every real β**, and at β = 0 exactly the product Haar measure over the slice. The space is a Hilbert space by Mathlib's Lp instances — positive by construction, with **no appeal to OS positivity**. The **vacuum** is the constant function 1, of norm 1 (`norm_osVacuum`); vacuum expectation equals the Gibbs expectation of the pulled-back observable (`vacuum_expectation`). Bounded slice observables act as multiplication operators (`sliceMul`), operator norm ≤ the bound, with **adjoint = pointwise conjugation** and real-valued ⇒ self-adjoint. What routes through 1b-B's open clause is stated **conditionally on it, by name**: the clause is packaged as `GramClause`, and `osForm_nonneg_of_gramClause` / `osForm_slice_nonneg_of_gramClause` assume it as an explicit hypothesis; positivity at β = 0 for slice observables is unconditional (`osForm_slice_zero`). One marginal identity (the pushforward of product Haar along the slice inclusion) is named, unproved, priced **S**, and **not used**.
- **Stage 1b-D — the transfer matrix: STANDS.** `LatticeTransfer.lean` (1,610 lines; 58 declarations, 13 definitions; build green, 3,551 jobs; eleven headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`). The transfer operator is the adjoint compression `T_β = A*† ∘L B*` on the OS space, where A* (`slicePull`) pulls a slice observable back along the time-0 restriction and B* (`nextPull`) does the same along the time-1 restriction transported to the time-0 slice; both legs are linear isometries into the same L^2 of the Gibbs measure, and both slice marginals are proved equal to the slice measure at every real β (translation invariance of the Gibbs measure, proved in the module in `Measure.map` form, by 1b-A's reflection argument run for translations). The operator package, at **every real β**: the **form identity** (`transferOp_inner`: ⟪g, T_β

f⟫ is the two-slice Gibbs kernel pairing, as an identity); **contraction** (`transferOp_norm_le`: ‖T_β‖ ≤ 1, by Cauchy–Schwarz against the two isometries); **vacuum fixed** (`transferOp_vacuum`: T_β Ω = Ω); **self-adjointness** (`transferOp_adjoint`, `transferOp_isSelfAdjoint`, at even N — from 1b-A's proved reflection invariance via the reflection step Ψ = (time shift by −2) ∘ θ, which exchanges the two slices; **no clause spent**); and **form positivity** (`transferOp_form_nonneg`: for pointwise real nonnegative slice observables, ⟪g, T_β f⟫ is real and nonnegative — proved from the positivity of the Gibbs measure alone, again with no appeal to OS-form positivity or the Gram clause). **The lattice gap at β = 0 is exact:** the kernel pairing factorizes over the disjoint slices (Stage 1a's β = 0 factorization), giving `transferOp_zero_apply : T₀ f = ⟪Ω, f⟫ • Ω` — **T₀ is exactly the orthogonal projection onto the vacuum line**, idempotent (`transferOp_zero_comp_self`), killing the vacuum-orthogonal complement; its spectrum is {1} on the vacuum line and {0} on the complement, so **the gap of T₀ is exactly 1**. That is the free (infinite-temperature) point; it is exact, and it is all the gap this stage claims.

Pitch-level shape after Stage 1b. Stages 0 and 1a stand whole; Stage 1b stands in A (structure), C (the space), and D (the operator package with the exact β = 0 gap), and in B it stands in the form, at β = 0, and as the proved Gram criterion, with general-β positivity the stage's single open clause. **PARTIAL-δ** — the scoping shape that would let this paper say "reflection positivity stands" for Stage 1b — is **NOT claimed**, and will not be claimable until 1b-B's clause closes. Stages 1c, 2, 3a/3b, 3c, and 4 are unbuilt and priced (§14). Nothing in this section takes a thermodynamic or continuum limit, constructs an interacting Poincaré representation, or closes any part of the Clay statement.

14. Frontier and priced windows

The following are named, framed, and priced — not claimed.

1. **Unbounded operator theory in Mathlib.** Densely defined closed operators, the unbounded adjoint, von Neumann's A** theorem, the Cayley transform, projection-valued measures, the unbounded spectral theorem, and Borel functional calculus are absent from Mathlib (surveyed 2026-10-11; five checkable window statements are filed in `SpectralFrontier.lean`). This is the boundary of formalized analysis on the spectral prong.
2. **Second quantization (bosonic) — discharged in Version 2.** The adjoint-relation assembly is discharged (§8B), and Γ(U) functoriality is discharged (§10, Task 3, closed by the Γ functor campaign of 2026-10-16): `Γ(1) = 1`, the diagonal action, multiplicativity, and unitarity are proved at finite modes. The **infinite-mode limit is now also discharged** (Version 3): the direct-limit Fock construction exists and the free theory stands at countably many modes (§10A, Pitch 2). What remains priced on this line: the **spectral** form of the free gap at countable modes (the eigenvalue-on-finite-particle-domain form is proved; Task 7 of the Pitch 2 scoping), rectangular functoriality beyond the padding identifications (Task 3b), interaction in the limit spaces (§10B's Route C:

FRONTIER), and the symmetric tensor-power unitary equivalence (blocked upstream, as priced at §8).

3. **Boost invariance** of the mass-shell measure — **DISCHARGED in Version 3** (§7 amendment): the one-dimensional Jacobian identity is proved and, with the factorization cap, the measure is invariant under the full restricted Lorentz group.
4. **Strong continuity** of the one-particle representation: a dominated convergence argument with the dominating function identified. Repriced by the parked-fleet gauntlet: the translation part is **S, discharge-ready** (the dominating function $4\|\psi\|^2$ and the dominated-convergence lemma are named in the source file); the Lorentz part is **S–M** (§7 amendment's discharge supplies the Jacobian domination). **Fock packaging** (the one-particle representation assembled on the Fock spaces of §§8–10A) is priced **M** by the concrete orthonormal-basis route; the abstract route stays killed (§15).
5. **n-point Wick expansion:** the Wick recursion at 2n points.
6. **Faithful SM gauge representation** (§11, Track 1): priced at the pre-Option-A representation or the manifold factor.
7. **Continuum limit and interacting 4D QFT.** No construction exists in this corpus. The Wightman verdicts of Rung 5 are toy-model verdicts; the general theory is not formalized.
8. **Yang–Mills existence and mass gap.** Quarantined T5. The Clay statement is formalized as a fixed target (`ClayStatement.lean`), and the dossier discipline records why finite-level gaps do not transfer: Osterwalder–Seiler's strong-coupling lattice gap excludes the continuum limit; Balaban's renormalization-group work is not a mass-gap proof; φ^4_4 triviality says nothing directly about Yang–Mills; spectral-gap undecidability results concern families of Hamiltonians, not the single Yang–Mills Hamiltonian.
9. **Bosonic essential self-adjointness** (§10B): the closure equality `hamPMap.closure = hamPMap†`, statable since the parked-fleet gauntlet found `LinearPMap.closure` in the pinned Mathlib. Priced **M–L** (restating the target, S; the domain argument, M–L), by the banded-truncation route (R2) or the Nelson-commutator route (R1). This is Pitch 3's sole remaining priced item.
10. **The `adag` repair and Yukawa.** The fermionic and bosonic Fock modules cannot be co-imported (both define `ThetLogos.adag`, with `ann`, `adag_basis`, `ann_basis` — the collision set is exactly those four names): a mixed fermion–boson (Yukawa) term cannot yet even be stated in one environment. The repair (rename the bosonic side, 82 use sites in 4 files, plus a co-import smoke module as the deliverable) is priced **S–M**; the Yukawa campaign on top of it is priced **M–L** (its walls fenced in advance: essential self-adjointness verbatim from item 9, and uniform persistence false by §10B's collapse counterexample).
11. **Reflection positivity at general β** (§13A, Stage 1b-B): the Gram/character-expansion factorization of the Wilson weight. Priced **XL, Mathlib-blocked** (needs Peter–Weyl orthogonality for general compact groups). Until it closes, PARTIAL-δ stays unclaimed.
12. **The lattice gap at general β** (§13A, Stage 1b-D): spectral input on the transfer operator at fixed $\beta > 0$ (classically, compactness of the continuous-

kernel operator and its spectrum). Priced **M–L**. A gap uniform in volume and coupling along a continuum trajectory is item 13's Stage 3c, not this.

13. **Pitch 4's unbuilt stages** (§13A): Stage 1c-0, a finite-order strong-coupling exhibit (M; its first-order plaquette coefficient in closed intertwiner form is a perimeter law at first order — any reading of it as area-law evidence is killed in advance); Stage 1c proper, uniform clustering (XL, killable — any clustering statement routing through gauge-variant intermediates is dead on arrival by §13A's Elitzur theorem); Stage 2, one exact blocking step (M–L; iterated flow XL, Balaban-scale); Stages 3a/3b, OS scaffolding (M / M–L); Stage 3c, uniform estimates along the continuum trajectory (research-scale, open in print); Stage 4, interacting Poincaré (FRONTIER).
14. **Schrödinger smalls** (§10C): the literal `NormedSpace.exp` identification of `evolOp` (S), and the Dyson first-order term (S–M: define and FTC-characterize, S; full Duhamel with remainder, M).
15. **Bridge follow-ons** (Appendix A): attainment of the Fiedler infimum (S), exponential decay at the Fiedler rate (M), and the Faraday storey (S–M; graph and lattice objects differ in kind — the lattice work of §13A does not cheapen it).
16. **Decoherence toy.** No partial trace exists in the corpus or the pinned Mathlib (grep-verified); a finite-dimensional toy built on a hand-rolled partial trace is priced **S–M** and is the program's cheapest fully standalone landing.

15. Kill ledger (selected)

Killed claims return only as new entries by proof; the originals stand.

- As-stated Connes–Kreimer antipode axiom — refuted; corrected version proved (different claim).
- ρ_M ⊗ ρ_F as product twist — does not typecheck.
- Strong Morley emergence (edge data determine 3+1) — T1 counterexample.
- p/n as a fundamental gauge-coupled ± pair; electron as an interaction product of that sector — killed by standard physics (isospin is an approximate global symmetry; lepton number, spin, and statistics all fail for the composite-electron reading). The honest survivors: approximate global isospin, the electron as a separate lepton field, beta decay as weak charged current.
- W5 monomial-range cyclicity as framed — false; span version proved.
- Bosonic-form W6 criterion for fermions — fermion-blind; Koszul version proved.
- "Finite triple converges" (continuum language) — killed; priced roadmap instead.
- The φ-tower (golden-ratio mass hierarchy) — killed against measured mass ratios.
- $\mathbb{C}^7$ torsion — killed; not resurrected by the Morley gauntlet.
- Uniform bosonic gap persistence for general quartic coupling — **disproved in-module** (§10B): attractive coupling gives eigenvalues (lam a + lam b)·n − n^2, unbounded below. The windowed statement is the true one and is proved.

- The weak-dichotomy exact-gap forms of both Pitch 3 campaigns — false by the two-zero-mode counterexample recorded in both modules; the strengthened (no-zero-mode) statements are proved as different claims.
- Positivity of the Osterwalder–Schrader form over observables supported on the reflection-fixed plane links — false (Stage 1b-B dossier's counterexamples: form value -2 at $\beta = 0$ over $\mathbb{Z}/3$; a U(1) character value $I_1(\beta)e^{i\gamma}$, not real, at fixed stratum variables). The strict-positive-half class is the true one.
- The abstract SymmetricAlgebra route to Fock packaging — killed (again, and permanently, in the 2026-10-17 gauntlet); the concrete orthonormal-basis port is the only route (§14, item 4).
- A separate Osterwalder–Schrader fit-audit intake (the Round 2 item) — killed as redundant: absorbed by Pitch 4's construction (§12 note). Its re-pinning half survives as a priced item, not as an audit.

16. What is NOT claimed (the immune system)

- **No 4D QFT.** The finite triple is zero-dimensional; the Wightman verdicts are one-mode toy verdicts; no interacting theory in four dimensions, or in any continuum, is constructed. (Interacting Hamiltonians at finite modes are constructed — §10B; the fence is stated at "No interacting continuum theory" below.)
- **No continuum limit.** The continuum is a priced roadmap throughout.
- **No mass gap.** The Clay target is fixed and quarantined; nothing here is a step in a Clay proof, only infrastructure such a proof would need. Parts of the official Clay Yang–Mills statement closed by this corpus: **0** — before and after every result in this paper.
- **No full SM gauge group from the finite representation.** The represented U(1) is provably not hypercharge (§11). The algebraic match holds at the level of the full finite algebra and its unimodular subgroup.
- **No derived masses, couplings, or mixing angles.** All are labeled inputs wherever they appear.
- **No physical predictions.** Spectral, thermal, and trace numbers in this corpus are properties of the mathematical models, not measurements of nature.
- **No derivation from analysis where axiomatization was used.** Rung 3's promotions and Rung 4's bridge axiom are labeled T2 at every citation.
- **The free-theory gaps are not the mass gap.** The derived gaps of §9 (Task 3), §10 (Task 2), and §10A (Pitch 2, at countably many modes) are statements about free Hamiltonians, with the one-particle gap Δ as an input; the persistence result of §10B is a statement about a finite-mode interacting Hamiltonian under its threshold; and the lattice gap of §13A is a statement about a finite-lattice transfer matrix at $\beta = 0$. None of them is the Yang–Mills mass gap, and none transfers to any interacting or continuum theory (§14, item 8).
- **No interacting continuum theory.** (Amended in Version 3; the Version 2 line "no interacting theory of either statistics" is overtaken by §10B and is withdrawn.) Interacting Hamiltonians at finite modes are constructed for both

statistics (§10B) — matter self-interaction only. No gauge-coupled interaction appears anywhere in this paper: a mixed fermion–boson (Yukawa) term cannot yet even be stated in one Lean environment, because the two Fock modules cannot be co-imported (the `adag` collision; repair priced, §14 item 10). No scattering content, no interacting Poincaré representation, and no interacting theory in the continuum is constructed or claimed.

- **No continuum Γ.** (Amended in Version 3; the Version 2 line "No infinite-mode or continuum Γ" was false from 2026-10-16 and is withdrawn.) Γ is verified at finite modes (§§9–10) and, for the free theory, at countably many modes (§10A). No continuum lift of Γ is proved or claimed.
- **No Poincaré representation on Fock space, and no interacting Poincaré representation.** (Amended in Version 3.) What is now proved is one-particle: the free mass-shell measure is invariant under the full restricted Lorentz group (§7 amendment). §9 (Task 4) discharges the Γ-lift's form at finite modes for mode-set-preserving groups only. No boost or continuum rotation is implemented as a strongly continuous unitary representation on any Fock space in this corpus (strong continuity and Fock packaging remain priced, §14 item 4), and the interacting continuum Poincaré representation is Pitch 4's Stage 4 — FRONTIER, unbuilt.
- **No reflection positivity at general coupling.** Stage 1b-B (§13A) proves the OS form, its positivity at $\beta = 0$, and the Gram criterion; positivity at general β is reduced to one named open clause (the Gram factorization of the Wilson weight, XL/Mathlib-blocked). **PARTIAL-δ is not claimed.** Nothing in §13A's Stages 1b-C/1b-D spends that clause: the OS space is positive by construction, and the transfer matrix's proved properties route through 1b-A's proved reflection invariance and the Gibbs measure's positivity alone.
- **No lattice gap at general coupling, and no uniformity.** The transfer matrix's gap is proved exactly at $\beta = 0$ only (§13A). At general β the gap is fenced and priced (§14, item 12); no gap uniform in volume or coupling is stated anywhere, and no continuum-limit conclusion is drawn from any finite-lattice statement in this paper (Stage 3c's uniform estimates are research-scale and open in print; Stage 4 is FRONTIER).
- **The bosonic interacting Hamiltonian is not called self-adjoint without its scoped senses.** Symmetry, sectorwise self-adjointness, and LinearPMap formal self-adjointness are proved (§10B); essential self-adjointness is priced (§14, item 9), and the unqualified headline is not spendable.
- **No bridge from the background algebra.** The modules of Appendix A (the $\mathbb{Z}_3$ and $\mathbb{Z}_{60}$ gradings, the flux-twist toy, the weld, and the graph-Laplacian bridge exhibit) are T1 algebra standing alone: no $\mathbb{Z}_3$ grading of the TRO, of any tripotent family, or of any Dirac operator is claimed, and no physical content is asserted for them.

17. Reproducibility

- **Source:** https://github.com/jeffgurd888/thet-logos (Lean sources in `lean/ThetLogos/`; verdict dossiers in `whitepaper/`).

- **Build:** with Lean 4 / Mathlib per `lean/lean-toolchain` and `lean/lake-manifest.json`: `cd lean && lake build` (green; covers the root-imported modules, and was green at 3,325 jobs at the Rung 9, Rung 8B, and Rung 10 landings). The remaining modules build individually: `lake build ThetLogos.<Module>` — the 2026-10-15 status per module is reported in §0, the Version 2 modules' targeted builds are reported in the §0 addendum, and the Version 3 modules' targeted builds (each green, with job counts) are reported in the §0 Version 3 addendum. The Version 3 inventory was taken at commit `4380a2a`.
- **Counts:** the counting method of §0 (Version 1–2 numbers) and of the §0 Version 3 addendum (all Version 3 numbers, one uniform method) reproduce every number in this paper from the sources.
- **Axioms:** `#print axioms <theorem>` in any module reproduces the footprints reported in §0.

Acknowledgments

This program formalizes. It did not originate the ideas, the problems, or most of the mathematics it verifies. The staircase this corpus stands on, named at the author's direction ("make sure the true credit goes to them"; the repository's `ACKNOWLEDGMENTS.md` carries the same text):

- **The problem.** Chen Ning Yang and Robert Mills — the gauge theory whose rigorous construction is the summit this program climbs toward. Arthur Jaffe and Edward Witten — the official Clay statement of the Yang–Mills existence and mass gap problem. The Clay Mathematics Institute — for posing it as a Millennium Problem.
- **The geometry.** Paul Dirac — the Dirac operator at the heart of every spectral triple in this corpus. Alain Connes — noncommutative geometry and the spectral triple; the distance formula, the spectral action, and the framework this corpus formalizes. Ali Chamseddine — co-architect of the spectral action and of the almost-commutative Standard Model geometry.
- **The modular theory.** Minoru Tomita and Masamichi Takesaki — the modular theory that turns a state into a flow. Alain Connes and Carlo Rovelli — the thermal time hypothesis.
- **The axioms and the fields.** Arthur Wightman — the axioms a rigorous quantum field theory must satisfy. Konrad Osterwalder and Robert Schrader — the Euclidean axioms and reconstruction. Werner Heisenberg — the operator language of quantum mechanics. Erwin Schrödinger — the equation whose dynamics this corpus formalizes. James Clerk Maxwell — the first gauge field theory. Albert Einstein — the field equations whose Hilbert-type terms the spectral action recovers at finite level. Isaac Newton and Gottfried Wilhelm Leibniz — the calculus and the limit: the continuum this program works to reach honestly.
- **The lattice.** Kenneth Wilson — lattice gauge theory and the renormalization group picture; the route §13A takes. Shmuel Elitzur — the theorem that gauge-variant quantities cannot acquire expectation values; proved here as a result,

not assumed as a slogan. Miroslav Fiedler — the variational eigenvalue bound used in this corpus's graph-Laplacian exhibit. Alfréd Haar — the measure every lattice expectation here is integrated against.
- **The analysis.** Laurent Schwartz — distributions and the function spaces of this corpus's analytic rungs. Marshall Stone and John von Neumann — the spectral and unitary machinery underneath the dynamics. Leonhard Euler — the action-and-extremum tradition the whole Lagrangian world descends from.
- **The staircase in modern form.** The Lean community and the Mathlib contributors — thousands of people whose verified library this corpus imports directly; our theorems stand on their theorems, mechanically and literally. The constructive field theory tradition — Glimm, Jaffe, Balaban, and many others — whose decades of work mark the frontier this program measures itself against.

Formalization, verification, and the finite-level results are ours. The concepts are theirs. If this corpus is ever cited, cite the originators first.

Appendix A. Background algebra (fenced — not part of the ladder)

This appendix reports five T1 modules that stand alone. They are **not** rungs of the ladder, **not** physics, and **not** claimed to connect to the corpus's operator-algebra structures; they are recorded here so the inventory is complete and the fence is explicit.

- **Ternary substrate (`TritGrading.lean`, 350 lines; T1).** A three-element type `Trit` (`neg | zero | pos`) with its full abelian- group laws proved by case analysis and a bundled additive equivalence `Trit.addEquivZMod3 : Trit ≃+ ZMod 3`; a submodule-valued grading structure `TritGrading` whose only axiom is multiplication closure, with commutator closure (`TritGrading.commutator_mem`) and adjoint- action closure (`TritGrading.ad_mem`) proved, plus a block-offset instance; honestly named commutator facts (`commutator`, `ad` — not "curvature", not "covariant derivative"); and the squared-commutator trace bound `trace_commutator_sq_nonpos` for Hermitian matrices (`trace [A,B]²` is not a Chern character, and is not named as one). Zero sorrys; headline axiom footprint exactly `[propext, Classical.choice, Quot.sound]`. **Connection audit (honest): none.** The corpus's "ternary" is the TRO product `⟨a, b, c⟩ = a · star b · c` — a 3-ary operation, a different notion of "three" from a cyclic grading; nothing in the corpus grades the TRO or its tripotents by $\mathbb{Z}_3$, and the corpus's operator-algebra gradings are $\mathbb{Z}_2$ (the spectral triple grading `γ_F`). The substrate stands alone as background algebra: hygiene and readiness, no forced bridges.
- **Sexagesimal clock (`SexagesimalClock.lean`, 433 lines; 31 theorem/lemma declarations, 12 definitions, 1 structure; T1).** The $\mathbb{Z}_{60}$ sibling of the trit module, built to the same standard: unique two-place sexagesimal representation for every n < 3600 (and the three-place version for n < 216000), both with uniqueness proved; the divisibility core (every k in {1,2,3,4,5,6} divides 60, and lcm(1,…,6) = 60 — the last depending on no axioms at all); the

canonical sub-clock reduction maps out of $\mathbb{Z}_{60}$, the mod-3 one named `toTritClock`; the Chinese remainder decomposition `crtEquiv : ZMod 60 ≃+* ZMod 3 × ZMod 4 × ZMod 5` with the projection compatibility `crtEquiv_fst` proved (the first component is the trit reduction); a `ClockGrading` structure with commutator and adjoint-action closure; and the headline `ClockGrading.reduceToTrit` — every $\mathbb{Z}_{60}$ clock grading pushes forward along the mod-3 reduction to a genuine `TritGrading`. That is grading bookkeeping: the trit clock sits inside the sexagesimal clock as gradings. Not a physical identification. Targeted build green (3,118 jobs); zero `sorry`, `admit`, or `native_decide`; eight headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`. One vocabulary fence, stated in the file header: the "modular" of this program is Tomita modular theory; this module is modular arithmetic, and the rhyme earns nothing.

- **Flux-twist toy (`FluxTwist.lean`, 582 lines; 38 theorem/lemma declarations, 10 definitions; T1).** A finite toy on the same shelf: the twisted shift on the ring of `Fin (n+1)` sites, the boundary bond carrying the flux phase. Proved: unitarity (`twistedShift_unitary`, both products with the conjugate transpose); flux periodicity (`twistedShift_add_two_pi`: $S(\theta + 2\pi) = S(\theta)$); gauge redistribution (`twistedShift_mul_gaugeMatrix` — the boundary twist is the uniform twist, conjugated by the diagonal gauge matrix whose site-k phase is $\theta \cdot k/(n+1)$); and the ring hearing the flux — the eigenphase $(\theta + 2\pi m)/(n+1)$, proved for the uniform twist on plain Fourier vectors and for the boundary twist on flux-carrying Fourier vectors (plain ones fail the wrap bond by exactly the flux factor; recorded in the file). The stretch target landed except its priced item: the hopping Hamiltonian is Hermitian (`hopping_isHermitian`) with eigenvalues $-\cos((\theta + 2\pi m)/(n+1))$; the Gibbs-state modular-flow generation statement is **priced, not formalized** (it needs matrix exponentials and the finite-dimensional modular theory of Gibbs states), and is stated as priced in the file header. The vocabulary fence is the module's point: "magnetism" appears only as flux twist — the Aharonov–Bohm effect in miniature. It is **not** a bridge to real magnetism or Maxwell's equations, and **not** a Yang–Mills rung. Targeted build green (3,120 jobs); zero `sorry`, `admit`, or `native_decide` in code; six headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`.
- **Weld duality (`WeldDuality.lean`, 893 lines; 73 theorem/lemma declarations, 18 definitions; T1).** Finite linear algebra: the finite shadow of the Morley idea of §13, built in both of its configurations as two faces of one module. The common core: for an involution η ($\eta^2 = 1$), the spectral projections `weldPlus` / `weldMinus` $= (1 \pm \eta)/2$ are complementary orthogonal idempotents, $\eta = P_+ - P_-$, the trace formula `tr P₊ = (n + tr η)/2` holds, and the signature count is rigid under conjugation. Face A is the grading split with a corrected involution: `corrInv` in parametrized form (an involution when $\eta^2 = A^2 = 1$, $\eta A + A\eta = 0$, $c^2 = 2$), and on the $\mathbb{C}^4$ arena ($\eta = \mathrm{diag}(1,1,1,-1)$) a corrected involution carrying a Hadamard block on the mixed directions (`etaCorr_sq`). Face B is the Dirac sign with the gap split: for $D = \mathrm{diag}(2,1,-1,-3)$, the sign involution η_D is Hermitian with $\eta_D^2 = 1$, its plus-projection is the positive spectral subspace by computation (`weldPlus_etaDirac`), and $H = D^2$ is split at $\Delta^2 = 2$ into complementary orthogonal projections with `gap_dichotomy` (every eigenvalue is 1 or ≥ 2). The linkers are proved, and they are what makes the two faces one

module: Face B is an instance of Face A (`signature_etaDirac`), so the signature count follows the **input** in both faces — the §13 kill (`signature_not_determined`) stands untouched, and nothing here derives (3,1); and the curved correction leaves the plus-projection's trace invariant while the projections genuinely differ (`trace_weldPlus_etaCorr`, `weldPlus_etaCorr_ne`) — the stability is not vacuity. The resting-place theorems the module was designed around are proved: the fixed set of an involution is exactly the range of its plus-projection (`fixed_iff_range_weldPlus`), and for two commuting involutions the joint fixed set is the range of the product projection (`joint_fixed_iff_range`). **Priced, not attempted:** the general correction-stability theorem (the plus-projection's count constant along a general correction, for arbitrary n) — only the worked-arena instance above is proved. The fence, stated in the file header: "2D holography" does not survive into this module — only the finite shadow does, dimension counted by traces of spectral projections. **Not** a Yang–Mills rung. `EdgeSignature.lean` is not imported (its §1 pair algebra is restated locally; disclosed in the file header). Targeted build green (3,120 jobs); zero `sorry`, `admit`, or `native_decide` in code; seven headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`.

- **Graph-Laplacian bridge exhibit (`BridgeLaplacian.lean`, 466 lines; 29 theorem/lemma declarations, 7 definitions; T1).** (Added in Version 3.) On a finite weighted graph (vertex type V, symmetric weights w as bare data — symmetry and nonnegativity are theorem hypotheses, never baked into the definitions): the bridge operator `dropVec F i j = F j − F i` between node potentials and edge drops, with its linearity, antisymmetry, and the telescoping path law whose closed-path corollary is the discrete Kirchhoff voltage law; Ohm's law as the middle factor `ohm w X i j = w i j * X i j`; and the **factorization** `laplacianOp_eq_diverge_ohm_drop` — $L = d^{T}\sigma d$ — a genuine theorem, not a definition (the divergence gathers transposed entries, so the two sides agree only after the symmetry of w is spent). With it: the current law ($\sum_i (LF)_i = 0$), the dissipation identity ($2\cdot\text{energy} = \sum_{ij} w\,i\,j\,(F_i - F_j)^2$) and energy nonnegativity, self-adjointness of the form, the kernel characterization (zero energy ⟺ F constant across every conducting edge; at all-positive weights, F constant), and the **Fiedler value** as a variational infimum with its bound proved whole (`fiedler_le_energy`, `fiedler_nonneg`) — the pre-authorized fallback of pricing the bound was not needed. The three readings of the exhibit (resistor network / graph heat / DeGroot consensus) are recorded as an equation identity — one operator equation read three ways — claiming nothing about real circuits, real heat, or real electorates. **Priced, not attempted:** attainment of the variational infimum (S), exponential decay at the Fiedler rate (M), and Faraday's law — the exhibit's second storey, needing cycle/flux data (S–M). Relation to Mathlib, stated in the dossier: Mathlib's `SimpleGraph.lapMatrix` is the unweighted 0/1 Laplacian; this exhibit differs in kind (arbitrary symmetric real weights, function-level operators, the explicit $d/\sigma/d^{T}$ factorization), and no novelty beyond that packaging is claimed. Not a Yang–Mills rung, and not connected to §13A's lattice: graph and lattice objects differ in kind, and no bridge between the exhibit and the lattice stages is claimed. Targeted build

green (3,117 jobs); zero `sorry`/`admit` in code; eight headline axiom checks exactly `[propext, Classical.choice, Quot.sound]`.

Notation. φ denotes the modular flux i[K, D] (Hermitian, `modularFlux_selfAdjoint`). Φ is reserved for the ontological glyph and is not used in this paper. Tet (the narrator of the companion books) and thet (the law) are distinct; neither is a mathematical object of this paper.